

RXNHAYSTACK: BEYOND NEEDLES IN LONG-CONTEXT STRUCTURED SCIENTIFIC DATA

Anonymous authors

Paper under double-blind review

ABSTRACT

Large language models (LLMs) deployed in AI-for-science pipelines increasingly need to reason over structured datasets that far exceed their context windows. Existing long-context benchmarks, however, rarely distinguish failures of dataset-scale access, systematic execution over structured objects, and domain-specific abstraction. We introduce RxnHaystack, a tiered benchmark for reasoning over large structured scientific datasets, instantiated in 122,456 cleaned USPTO reactions. Its 100 questions progress from exact retrieval and deterministic aggregation to reaction-level inference and relational reasoning over reaction graphs. We evaluate three access and execution interfaces including plain LLM prompting, CodeAct, and Recursive Language Models (RLMs), with six models spanning several model families and providers. Across 21,500 question-level trajectories in terminal experimental arms, the performance of plain LLM prompting declines consistently as the accessible corpus grows from 100 to 500 reactions, while CodeAct scaling depends on the model and task. Both interfaces can operate only on the portion of the corpus supplied in their prompt. RLMs additionally provide recursive access to the complete 122,456-reaction corpus, allowing us to test what remains difficult when the full dataset is accessible. Across Qwen3.5-397B-A17B, DeepSeek-V4-Flash-0731, Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5, mechanically specified graph operations average 0.81 macro-F1, yet mechanism classification averages 0.34 and route and multi-constraint tasks average 0.08. Fixed-cardinality contexts and chemist-validated executable-rule controls further separate the effects of corpus scale, chemistry-rule induction, and faithful execution. RxnHaystack therefore turns failures of large-scale access, structured execution, chemical abstraction, and multi-step reasoning into separately measurable phenomena.

1 INTRODUCTION

Frontier LLMs now support context windows of a million tokens or more (Comanici et al., 2025), yet substantial reliability gaps remain. Retrieval accuracy falls when evidence appears in the middle of a long document (Liu et al., 2023); attention requires special handling over long sequences (Xiao et al., 2024); and models struggle to combine information spread across a large context even when each fact is individually accessible (Hsieh et al., 2024).

Scientific work introduces an additional challenge: many questions require a systematic operation over structured records. A biologist may compare genetic perturbations across cell states and experimental conditions (Dixit et al., 2016; Peidli et al., 2024); a materials scientist may filter structures by several composition and property constraints (Jain et al., 2013); and an adaptive experiment may aggregate a time series before selecting the next measurement (King et al., 2004; Kuroki et al., 2026). These analyses involve filtering, joining, aggregating, or traversing records, together with a domain rule that determines which records match or how they are linked. A textual encoding makes the records readable by an LLM, but reliable analysis still depends on executing the operation systematically and choosing a scientifically valid rule.

A benchmark for this setting requires four ingredients: a corpus exceeding ordinary prompt budgets, exact relations between records, tasks that vary independently in operational and domain demands, and ground truth auditable without the model under test. Reaction chemistry supplies all four within a single representation. Reaction SMILES compactly encode molecules and transformations;

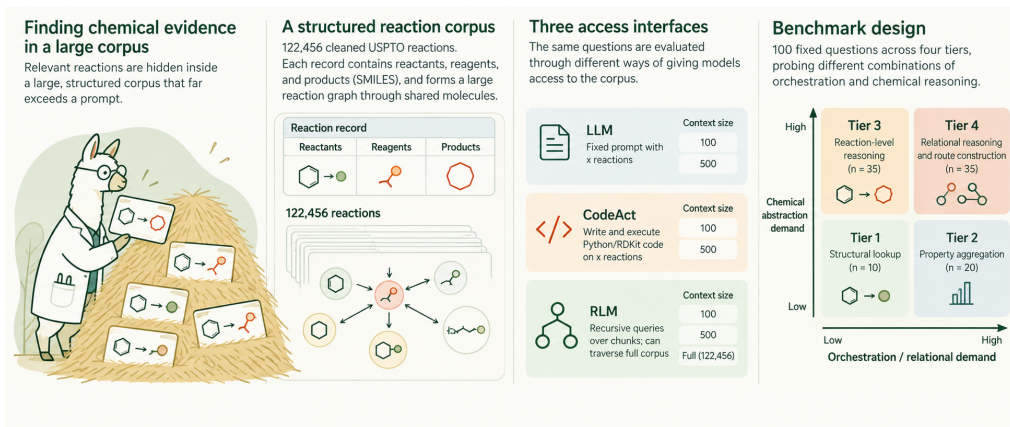


Figure 1: Overview of RxnHaystack. Models answer the same 100 questions over a structured corpus of 122,456 reactions through plain prompting, CodeAct, or RLM access. The four benchmark tiers vary the demands placed on chemical abstraction and structured orchestration.

canonical molecular identity links products to downstream reactants, forming a synthesis graph; and cheminformatics tools can verify molecular properties, bond changes, reaction patterns, and graph relations. The USPTO 2023 corpus supplies realistic scale and chemical diversity (van der Lingén, 2023). It therefore supports controlled experiments that range from exhaustive filtering to mechanism classification and multi-step route construction within one data representation and scoring framework.

These operations map directly onto recurring work in reaction informatics. Exact and similarity-based searches retrieve precedents from patent and laboratory reaction collections; corpus-wide filters and transformation labels support database curation, quality control, and reaction classification; and synthesis-graph queries recover connected sequences and candidate routes. Reaction fingerprints are designed for search and categorization (Schneider et al., 2015; Probst et al., 2022), structured reaction databases prioritize validation and reuse (Kearnes et al., 2021), and computer-aided synthesis systems organize known transformations into routes (Segler et al., 2018; Genheden et al., 2020). Reliable execution of these operations could assist cheminformaticians who maintain reaction collections, analyze patent chemistry, or provide precedent and route analyses to medicinal and process chemists.

We introduce RXNHAYSTACK, a benchmark that characterizes each question along two dimensions while varying dataset scale independently. *Operation/orchestration demand* captures the structured computation needed to scan, aggregate, link, or traverse the records. *Chemistry-abstraction demand* captures the knowledge needed to formulate the relevant structural, mechanistic, or route-level rule. The benchmark tiers sample different combinations of these demands. For example, molecular-weight filtering and mechanism classification both require a corpus-wide scan, while the latter also requires a chemical account of the transformation. Graph traversal creates the opposite contrast: it can require extensive orchestration even when every linking rule is stated explicitly.

Our contributions are:

- We construct 100 questions over 122,456 cleaned patent reactions, spanning exact lookup, deterministic aggregation, reaction classification, and constrained graph search. Section 3 defines the task dimensions, context construction, and deterministic evaluation.
- We compare plain prompting, CodeAct (Wang et al., 2024), and Recursive Language Models (RLMs) (Zhang et al., 2026) across model families, context scales, and task groups. Section 4 reports both aggregate performance and task-level heterogeneity.
- We introduce controlled interventions on answer cardinality and access to executable chemistry rules. These experiments distinguish failures caused by corpus scale, chemistry-rule induction, and faithful execution (Section 4.2).
- We preserve task generators, scorers, execution traces, resource measurements, and reconstruction instructions so that every reported result can be audited and extended (Appendix E).

2 RELATED WORK

Long-context evaluation. SCROLLS (Shaham et al., 2022), LongBench (Bai et al., 2025), and L-Eval (An et al., 2023) assemble long-document understanding tasks, while RULER (Hsieh et al., 2024) and InfiniteBench (Zhang et al., 2024) stress retrieval, aggregation, and multi-step inference at larger scales. These benchmarks establish that nominal context length does not guarantee reliable use of distant information. Their inputs are predominantly natural-language sequences. RxnHaystack complements them with a database-like scientific setting in which success requires explicit operations over molecular records and relations.

Molecular and reaction benchmarks. MoleculeNet evaluates molecular property prediction (Wu et al., 2018); USPTO-derived tasks support reaction-outcome prediction and retrosynthetic planning (Schwaller et al., 2019; Segler et al., 2018); and other benchmarks measure reaction yield (Ahneman et al., 2018) or broad chemistry knowledge in language models (Mirza et al., 2025). These resources have driven substantial progress on molecular representations and chemical prediction. Their standard instances provide a molecule, reaction, or short problem as the input. RxnHaystack instead asks a model to operate over a shared reaction corpus, making dataset access, exhaustive execution, and chemical abstraction visible within the same evaluation.

Retrieval, tools, and recursive access. Retrieval-augmented generation (RAG) selects a compact evidence set before generation (Lewis et al., 2021); reaction fingerprints such as DRFP provide a domain-specific similarity representation for chemical retrieval (Probst et al., 2022). ReAct interleaves language-model reasoning with tool use (Yao et al., 2023), and CodeAct uses executable programs as the action space (Wang et al., 2024). RLMs expose an oversized context through recursive sub-calls (Zhang et al., 2026). These approaches offer different mechanisms for selecting, accessing, and processing external information. RxnHaystack holds the scientific questions and scoring rules fixed while comparing representative interfaces for corpus access and structured execution.

3 BENCHMARK

3.1 DATASET

All tasks are grounded in the USPTO 2023 dataset (van der Ling, 2023) of 137,261 organic reactions from US patent filings. A cleaning pipeline discards malformed lines, rejects reactions with molecules that fail RDKit sanitization, and removes duplicates under the canonicalized (reactants, reagents, products) triple, leaving 122,456 reactions. Each reaction is one line of the form `index reactants>reagents>products` with dot-separated canonical SMILES fields. The dataset spans amide bond formation, palladium-catalysed cross-couplings, protecting-group chemistry, ring-forming reactions, and functional-group interconversions, giving realistic chemical diversity.

3.2 TIER DESIGN

We characterize tasks along two dimensions and evaluate them across a separate experimental scale axis. *Operation/orchestration demand* is the computation a model must coordinate across entries once the task-relevant rule is available. *Chemistry-abstraction demand* is the domain knowledge needed to formulate the correct structural, mechanistic, or route-level rule. Context size controls dataset scale: the amount of the reaction corpus the model must access and traverse. This decomposition identifies potential failure sources without assuming that the two demands combine multiplicatively. Related task groups provide internal controls: exact lookup stresses exhaustive access, descriptor filters add corpus-wide computation with shallow chemical abstraction, and reaction classification retains the scan structure while deepening the required chemistry. Likewise, path queries over the synthesis graph contrast purely structural traversal with chemically constrained routes. The task dimensions apply at subgroup granularity, and tier labels are coarse bundles whose table ratings denote their maximum demand. Table 1 summarizes the design; full definitions appear in Appendix A. The 100 questions are fixed across runs with deterministic RDKit-computed ground truth; randomness enters only through context sampling.

Table 1: Benchmark tiers. Q = number of questions; OO/CA give the maximum operation/orchestration and chemistry-abstraction demand within each coarse tier bundle. Dataset scale is varied independently through context construction.

Tier	Q	OO	CA	Representative tasks
1 Structural lookup	10	low	low	Find all reactions producing a given canonical SMILES.
2 Property aggregation	20	high	low	Filter by molecular-weight change, ring-count change, aromatic ring formation, and their conjunctions.
3 Reaction-level reasoning	35	high	high	Classify by bond changes, scaffolds, reagents, stereochemistry, SMIRKS functional-group templates, and mechanism templates.
4 Relational reasoning	35	high	high	Enumerate synthesis chains, hub molecules, protecting-group pairs, truncated route prefixes, SMIRKS-constrained 2-step chains.

Tier 1 (10 questions) is exact-match retrieval: given a target product SMILES, return all reaction indices whose product side contains it after canonicalization. No chemistry knowledge is required; the challenge is accurate scanning of a large structured corpus.

Tier 2 (20 questions) requires computing a molecular descriptor for every reaction in context and filtering by threshold: molecular-weight increase beyond τ Da, ring-count increase of exactly δ , formation of exactly δ new aromatic rings, and conjunctions of weight and ring criteria. The chemistry is shallow but the computation is non-trivial at scale.

Tier 3 (35 questions) requires classifying individual reactions by chemical content. This is the work of a trained organic chemist reading a reaction table: recognizing what bonds have formed or broken, which functional groups have been transformed, what the reagents imply about the mechanism, and whether stereochemistry has been set or preserved. Tasks fall into six sub-categories of increasing depth: bond-level connectivity changes (3Q), fused-heterocycle and ring-construction scaffolds (3Q), reagent classes such as transition-metal catalysts and amide-activating agents (2Q), de novo stereocenter creation and E-alkene formation (2Q), functional-group transformations defined by Rxn-INSIGHT SMIRKS templates (Dobbelaere et al., 2024) covering amide couplings, protecting-group manipulations, and named cross-couplings (15Q), and mechanism-level templates from a mechanistic reaction dataset (Joung, 2025) covering pathways such as Wittig olefination, Mitsunobu, and Staudinger reduction (10Q).

Tier 4 (35 questions) shifts to reasoning over the network of reactions, where chains are linked by exact canonical-SMILES equality between a product of step i and a reactant of step $i + 1$. This mirrors the logic of multi-step synthesis planning, where intermediates flow from one reaction to the next, and requires the model to hold a graph structure in mind across all reactions in context, which is qualitatively different from any per-reaction classification. Sub-groups: mechanical graph tasks covering fixed-length and longest-chain enumeration and hub-molecule identification (5Q); chemically constrained chains covering functional-group transformation chains, protecting-group install/remove pairs, and ring-construction chains (10Q); and two substantially harder groups (20Q): *truncated synthesis routes* (T4.16–T4.25; Appendix A), where the final step of a 5-reaction route to a named target is withheld, requiring the model to identify prefixes compatible with an unseen terminal step; and *SMIRKS-constrained 2-step chains*, requiring two independent reaction-template matches plus a cross-chain chemical-persistence condition simultaneously. The truncated-route questions identify each target by name without providing its structure, so they jointly test name-to-structure interpretation and route-prefix recovery.

3.3 EVALUATION SETUP

Ground truth construction. For each question, the ground truth is the complete set of correct answers over the full dataset: reaction indices for Tiers 1–3 and task-specific chains, chain indices, reaction pairs, or molecules for Tier 4. All ground-truth sets are precomputed offline using RDKit

chemistry rules and exact canonical-SMILES rules for linking reactions into chains, then stored with each task, so evaluation is deterministic and independent of any inference-time chemistry tooling.

Context construction. Each question is presented with a context of x reaction lines. For index-return tasks, we deterministically select a subset of correct reactions in proportion to $|G|x/N$, subject to a small minimum. For relational tasks, we select complete answers under the analogous budget and include every supporting reaction. Random non-answer reactions fill the remaining positions. Evaluation uses the resulting in-context ground truth G_c .

This construction sets positive recall to one for each sampled instance: every answer that will be scored is already present in the context. Replacing the sampler with a reaction retriever such as DRFP (Probst et al., 2022) could change the accompanying negatives or supporting records, but it could not improve recall of the fixed set G_c . The sampled conditions therefore measure recognition, structured processing, and exhaustive enumeration after candidate selection. Retrieval quality and the choice of supporting records require a separate retrieval experiment.

Answer format and metrics. Models return all satisfying task-specific items, or -1 if none. Each question is scored as set retrieval: precision, recall, and F1 of the predicted set against G_c (zero when denominators vanish), plus exact-set-match as a binary indicator. Tier-level and overall results are macro-averaged over questions, treating each question equally since answer-set sizes range from single digits to thousands.

Human baseline. We collected independent first-submission answers from three anonymous chemistry-aware volunteers. Annotators could use RDKit, scripts, notebooks, literature or web search, language models, coding agents, offline processing, or consultation, but were instructed to inspect the method and make the final judgment themselves. Ground truth and evaluative feedback remained hidden until submission. Assignments were stratified across tiers and task families with intentional overlap rather than requiring every annotator to complete the full suite. We retain exact submitted values and revision histories, and rescore first submissions with order-invariant set comparison against the corrected, versioned canonical answers. Abstentions and missing assignments do not enter accuracy metrics. Full study details appear in Appendix D.

Inference paradigms. Three paradigms receive identical contexts and prompts. *Plain LLM*: the full context and question in a single prompt, answered in one forward pass without code. *CodeAct* (Wang et al., 2024): an agent that iteratively writes and executes Python (with RDKit) over the context lines. *RLM* (Zhang et al., 2026): recursive decomposition of the context into sub-calls over smaller windows, the only evaluated paradigm able to query the entire 122,456-reaction dataset; for RLM the context-size parameter controls the per-level chunk size. RLM supplies a full-corpus access mechanism, but access alone does not ensure exhaustive or chemically valid processing. These paradigms probe different access and execution mechanisms; the benchmark tasks do not depend on any one of them.

Models. We evaluate six checkpoints spanning open- and closed-weight model families: DeepSeek-V4-Flash-0731 (DeepSeek-AI, 2026); Qwen3.5-397B-A17B (Qwen Team, 2026); GLM-5.2 (Z.ai, 2026); Google Gemini 3.7 Flash (Google, 2026); OpenAI GPT-5 mini (OpenAI, 2025); and Anthropic Claude Haiku 4.5 (Anthropic, 2025). Cross-interface comparisons use the five models with terminal relevant arms; completed GLM-5.2 conditions remain included in interface-specific summaries.

4 RESULTS

We first examine corpus scaling across model families. For LLM and CodeAct, x is the number of reaction lines in the prompt; for RLM it is the recursive chunk size, and "full" permits traversal of all 122,456 reactions. Figure 2 averages the terminal core-benchmark arms for each interface. Plain LLM prompting degrades as the supplied corpus grows, executable access remains strong on deterministic operations, and RLM retains partial performance on the chemistry-intensive and relational tiers at full-corpus scale. The task-level analysis complements these cross-model trends with a fixed-model view of GPT-5 mini over five runs and all 100 questions in every core condition.

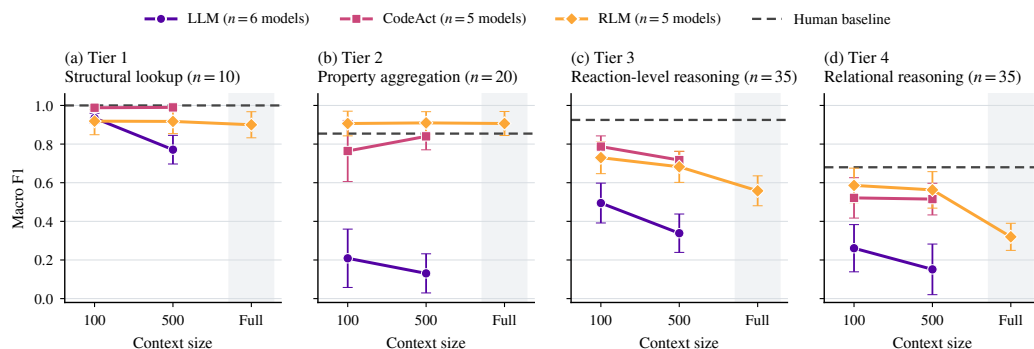


Figure 2: Cross-model scaling by tier. Points are unweighted means across terminal model arms (LLM: six models; CodeAct: five; RLM: five), and error bars show standard errors across model-level means. Scores are question-weighted within each model, and terminal failures contribute zero. Dashed horizontal lines show the human macro-F1 reference on assigned full-corpus questions. The shaded full-corpus setting is RLM-only. Because $|G_c|$ grows with x , slopes reflect both context and required answer-set size.

Tier 1: Structural lookup. At $x = 100$, plain LLM prompting, CodeAct, and RLM score 0.935, 0.988, and 0.919. At $x = 500$, the corresponding scores are 0.771, 0.990, and 0.917; full-corpus RLM reaches 0.900.

Tier 2: Property aggregation. This tier most clearly separates prompting from executable computation. Plain LLM prompting scores 0.209 at $x = 100$ and 0.131 at 500. CodeAct reaches 0.764 and 0.840, while RLM scores 0.906, 0.910, and 0.907 at $x = 100$, 500, and full-corpus scale. The descriptor filters become substantially more tractable once the interface supports systematic execution.

Tier 3: Reaction-level reasoning. RLM scores 0.730 at $x = 100$, 0.682 at 500, and 0.558 at full scale. Plain LLM prompting falls from 0.495 to 0.339, while CodeAct changes from 0.787 to 0.717. The task-level decomposition (Appendix 31) reveals substantial variation beneath these tier means: full-scale RLM reaches 1.00 on bond-level operations and 0.99 on stereochemistry, but falls to 0.51 on functional-group operations and 0.33 on mechanisms. Because the required answer set grows with x , cross-scale differences alone do not identify the source of this degradation; Section 4.2 introduces controlled comparisons.

Tier 4: Relational reasoning. RLM changes from 0.586 ($x = 100$) to 0.563 (500) and 0.320 (full), while plain LLM prompting changes from 0.261 to 0.151 and CodeAct remains near 0.52 across the two sampled contexts. At full scale, RLM reaches 0.960 on mechanical graph tasks, 0.459 across the ten chemically constrained questions, and 0.045 on route and multi-constraint tasks in the fixed-model analysis; truncated-route recovery reaches 0.012 (Appendix 31). Cardinality explains only part of this separation: prospective sets average 30.7 scored items versus 979.6 for chemically constrained tasks despite much lower F1 (Appendix A.5). Full-corpus Tier 4 runs were capped at 30 GB RAM. The large gap under a common memory ceiling, together with heuristic sampling in sequential-chain traces, points to search and chemical abstraction as the principal remaining challenges.

Human baseline. Three chemistry-aware volunteers answered stratified subsets of full-corpus questions with unrestricted tool use and human verification. Across 51 non-abstained first submissions, macro F1 was 0.842 and exact-set accuracy was 0.686. Macro F1 by tier was 1.000, 0.854, 0.925, and 0.680. Figure 2 includes these per-tier values as reference lines; Appendix D describes the assignment design, timing, tool use, corrected-ground-truth rescoring, and agreement analysis.

Observed failure modes. Representative traces expose chemistry-specific errors alongside computational ones. CodeAct misuses `RunReactants` as a class-membership test, scoring zero on

Table 2: Full-corpus RLM capability split across Qwen3.5-397B-A17B, DeepSeek-V4-Flash-0731, Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5. Means are unweighted across models after question-weighted aggregation within each model; ranges show model heterogeneity.

Operation group	Mean F1	Model range
Bond changes	0.852	0.611–1.000
Stereochemistry	0.827	0.586–0.998
Mechanism classification	0.344	0.207–0.571
Mechanical graph operations	0.806	0.395–1.000
Chemically constrained graph operations	0.564	0.293–0.933
Route and multi-constraint tasks	0.076	0.020–0.250

acyl-chloride coupling; at $x = 500$, RLM replaces structural SMIRKS with reagent-string heuristics, yielding 116 false positives for two ground-truth chains. On truncated routes, RLM counts n characters and halogen tokens in SMILES to guess a withheld target, while CodeAct applies the same target-agnostic chemical test to every target. Appendix C provides the detailed trace-based analysis and the corresponding chemical corrections.

Accuracy–cost trade-off. Across Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5, the relative cost of executable access depends on the task tier. At $x = 100$, CodeAct averages \$0.009–\$0.042 per question trajectory across the four tiers, while RLM averages \$0.018–\$0.088. On Tier 4 at $x = 500$, RLM reaches 0.578 F1 at \$0.108 per trajectory and CodeAct reaches 0.553 at \$0.076; full-corpus RLM reaches 0.332 at \$0.168. Figure 14 shows the aggregate trade-off, and Figure 15 separates the three models.

4.1 DOES THE CAPABILITY SPLIT GENERALIZE ACROSS MODELS?

We test whether mechanically specified graph operations are consistently easier than chemically intensive tasks across models by evaluating Qwen3.5-397B-A17B, DeepSeek-V4-Flash-0731, Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5 (Table 2), scoring unsuccessful trajectories as zero.

The ordering persists across all five model families. Bond-change and stereochemistry operations reach mean F1 scores of 0.852 and 0.827, while mechanism classification averages 0.344. The separation is sharper in Tier 4: mechanically specified graph operations average 0.806, chemically constrained graph operations 0.564, and route or multi-constraint tasks 0.076. Mechanical graph performance spans 0.395–1.000, yet each model scores higher on these operations than on mechanism classification and route or multi-constraint tasks. The cross-model evidence therefore preserves the capability split while narrowing its interpretation: explicit graph procedures are generally easier to execute than chemically intensive classification and route construction, with substantial variation within every operation class.

4.2 IS THE BOTTLENECK SCALE, CHEMICAL ABSTRACTION, OR EXECUTION?

Aggregate scores alone cannot identify which task demand caused a failure, so we use controlled interventions to remove specific alternative explanations. The preceding scaling curves change both corpus size N and the number of positive reactions K . We isolate these factors on the 65 Tier 1–3 questions whose answers are individual reaction rows. For GPT-5 mini RLM, we construct contexts containing exactly K positives and exclude every other positive. At fixed $K = 1$, Tier 2 remains between 0.961 and 1.000 as N grows from 100 to the full corpus, whereas Tier 3 falls from 0.671 to 0.222 (Figure 3a). Conversely, at fixed $N = 5,000$, increasing K from 1 to 20 leaves Tier 2 unchanged (0.990 to 0.981) and improves Tier 3 from 0.344 to 0.525 (Figure 3b). The Tier-3 scale degradation therefore persists independently of positive-answer cardinality in this controlled subset.

We next intervene on whether the model must infer the chemistry rule. For five task configurations (16 questions), we give the RLM an answer-free, executable chemistry rule: a short Python/RDKit function that maps a reaction or molecule to the task-relevant chemical feature, for example, whether a reaction forms an amide bond. Each rule is derived from the benchmark’s chemist-validated

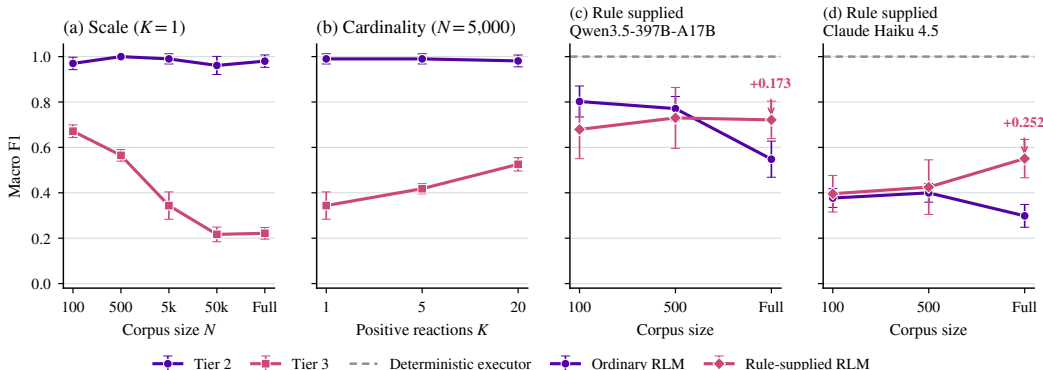


Figure 3: Controlled separation of scale, answer cardinality, chemistry-rule inference, and execution. (a) GPT-5 mini RLM with exactly one positive reaction while corpus size grows. (b) The same model at fixed $N = 5,000$ while positive cardinality varies. (c–d) Ordinary and rule-supplied RLM on 16 questions, compared with deterministic exhaustive execution of the same validated chemistry rules. Points are question-weighted means over five runs; error bars show one run-level standard deviation.

ground-truth computation and exposes only the per-record feature calculation; answer indices, answer counts, precomputed matches, graph search, and path assembly remain hidden. At full-corpus scale, supplying these rules raises Qwen3.5-397B-A17B from 0.548 to 0.721 and Claude Haiku 4.5 from 0.298 to 0.551 (Figure 3c–d). The aggregate gains are driven by the reaction-class and mechanism tasks. Qwen3.5-397B-A17B’s stereochemistry score falls under the supplied rule, and Tier-4 graph effects are mixed (Appendix A.7). Deterministic exhaustive execution of the same rules reaches 1.000 on every task, while rule-supplied RLM remains imperfect. Thus corpus scale, inference of the appropriate chemistry rule, and faithful application of that rule constitute distinct bottlenecks; describing them as one undifferentiated “reasoning” failure obscures the observed error structure.

4.3 WHAT INFORMATION ENABLES TRUNCATED-ROUTE RECOVERY?

Task 16 jointly requires target identification, inference of a plausible final transformation, and recovery of a compatible multi-step prefix from the reaction graph. We separate these capabilities using three target molecules and three prompt conditions: target name, canonical target SMILES, and canonical target SMILES with a chemist-validated description of the withheld final transformation. Every reaction producing the exact target is removed from the searchable corpus, preventing direct target lookup. Qwen3.5-397B-A17B and Claude Haiku 4.5 receive full-corpus RLM access for five repetitions; all conditions share the same targets, ground truth, and corpus construction.

Supplying the target structure alone yields zero exact-route recovery for both models (Table 3). A broad final-transformation class provides a substantial signal, raising Qwen3.5-397B-A17B from 0 to 0.378 macro-F1 and Claude Haiku 4.5 from 0 to 0.133. The effect is target-dependent: Qwen3.5-397B-A17B improves on two of the three targets and Claude Haiku 4.5 on one, while neither model recovers the pyrimidine–piperazine route under the structure-plus-class condition (Appendix A.8). The remaining gap is large even though the model receives both the target structure and the final-step class. This locates the difficulty at two successive stages: choosing the terminal chemical transformation and using that abstraction to find and assemble compatible preceding reactions. The gap between mechanical graph operations and route construction therefore extends beyond name-to-structure knowledge.

Table 3: Task-16 truncated-route control. Values are mean macro-F1 $\pm$ one run-level standard deviation over three targets and five repetitions. Every reaction producing the exact target is excluded from the searchable corpus.

Model	Target name	Target structure	Structure + final-step class
Qwen3.5-397B-A17B	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.378 $\pm$ 0.169
Claude Haiku 4.5	0.052 $\pm$ 0.117	0.000 $\pm$ 0.000	0.133 $\pm$ 0.183

5 CONCLUSION

We introduced a tiered benchmark over 122,456 USPTO reactions for diagnosing reasoning over large structured scientific data. Across Qwen3.5-397B-A17B, DeepSeek-V4-Flash-0731, Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5, mechanically specified graph operations remain substantially easier than mechanism classification and route/multi-constraint tasks, although no operation is universally solved across models. Matched-cardinality results show that Tier-3 degradation persists as corpus size grows with one positive held fixed, and chemistry-rule controls separate failure to infer the correct local rule from failure to apply it exhaustively. The evidence therefore identifies distinct bottlenecks in dataset access, chemistry-rule inference, and structured execution; a single broad “chemical reasoning” deficit cannot account for the observed pattern.

Reliable performance on these tasks would support exhaustive reaction-database filtering and curation, consistent annotation of transformations and mechanisms, protecting-group analysis, construction and traversal of synthesis graphs, and retrieval of routes satisfying explicit chemical constraints (Kearnes et al., 2021; Schneider et al., 2015; Probst et al., 2022; Segler et al., 2018; Genheden et al., 2020). The truncated-route control further tests recovery of corpus-supported prefixes after every direct target-producing reaction has been removed. These operations are useful components of reaction informatics and computational synthesis workflows. They evaluate corpus-supported analysis and route recovery, not experimental feasibility, yield, selectivity, safety, or successful laboratory synthesis.

Reaction chemistry makes this decomposition independently measurable through compact structural representations, deterministic cheminformatics operations, and exact relations between records. Other scientific environments motivate the same design question when an agent must traverse structured observations, apply a systematic operation, and infer the domain abstraction that defines a valid result. Whether the observed failure structure transfers to those environments remains an empirical question. RxnHaystack establishes the diagnostic framework in reaction chemistry, where its factors can be controlled and its answers can be checked exactly.

AI USE STATEMENT

Generative AI tools were used to assist with software development, experiment orchestration and debugging, aggregation and visualization of experimental results, and language editing. The authors reviewed AI-assisted code, analyses, figures, and text; verified software changes with automated tests and experimental artifacts; and take responsibility for the final content of this work, including all claims and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The anonymous supplement includes all 100 questions and reference answers, answer-generation code, context builders and seeds, scorers, experiment definitions, dataset reconstruction instructions and file fingerprints, and a version-pinned software specification. The code and benchmark will be released publicly upon acceptance. Appendix E describes the release and contamination policy. The appendix also reports model calls, token use, latency, wall time, memory, and recorded API charges in USD by tier and interface; models accessed through institutional inference endpoints are identified separately from paid API use.

REFERENCES

- Derek T. Ahneman, Jesús G. Estrada, Shishi Lin, Spencer D. Dreher, and Abigail G. Doyle. Predicting reaction performance in c–n cross-coupling using machine learning. *Science*, 360(6385):186–190, 2018. doi: 10.1126/science.aar5169.
- Chenxin An, Shansan Gong, Ming Zhong, Xingjian Zhao, Mukai Li, Jun Zhang, Lingpeng Kong, and Xipeng Qiu. L-eval: Instituting standardized evaluation for long context language models. *arXiv preprint arXiv:2307.11088*, 2023.
- Anthropic. Claude Haiku 4.5 system card. <https://www-cdn.anthropic.com/7aad69bf12627d42234e01ee7c36305dc2f6a970.pdf>, 2025. Accessed 2026-09-24.
- Yushi Bai, Shangqing Tu, Jiajie Zhang, Hao Peng, Xiaozhi Wang, Xin Lv, Shulin Cao, Jiazheng Xu, Lei Hou, Yuxiao Dong, Yizhi Zhang, Jie Tang, and Juanzi Li. Longbench v2: Towards deeper understanding and reasoning on realistic long-context multitasks. *arXiv preprint arXiv:2412.15204*, 2025.
- Sebastian Bordt, Suraj Srinivas, Valentyn Boreiko, and Ulrike von Luxburg. How much can we forget about data contamination? In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 4998–5016, 2025. URL <https://proceedings.mlr.press/v267/bordt25a.html>.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- DeepSeek-AI. DeepSeek-V4-Flash update. <https://api-docs.deepseek.com/updates/>, 2026. API documentation update dated 2026-07-31; accessed 2026-09-24.
- Atray Dixit, Oren Parnas, Biyu Li, Jenny Chen, Charles P. Fulco, Livnat Jerby-Arnon, Nemanja D. Marjanovic, Danielle Dionne, Tyler Burks, Raktima Raychowdhury, Britt Adamson, Thomas M. Norman, Eric S. Lander, Jonathan S. Weissman, Nir Friedman, and Aviv Regev. Perturb-seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell*, 167(7):1853–1866.e17, 2016. doi: 10.1016/j.cell.2016.11.038.
- Michael R. Dobbelaere, Istvan Lengyel, Christian V. Stevens, and Kevin M. Van Geem. Rxn-INSIGHT: Fast chemical reaction analysis using bond-electron matrices. *Journal of Cheminformatics*, 16(1), 2024.
- Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12): 86–92, 2021. doi: 10.1145/3458723.
- Samuel Genheden, Amol Thakkar, Veronika Chadimová, Jean-Louis Reymond, Ola Engkvist, and Esben Bjerrum. AiZynthFinder: A fast, robust and flexible open-source software for retrosynthetic planning. *Journal of Cheminformatics*, 12(1):70, 2020. doi: 10.1186/s13321-020-00472-1.
- Google. Introducing Gemini 3.7 Flash. <https://blog.google/innovation-and-ai/models-and-research/gemini-models/introducing-gemini-3-7-flash/>, 2026. Accessed 2026-09-24.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024.
- Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, and Kristin A. Persson. Commentary: The materials project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1(1):011002, 2013. doi: 10.1063/1.4812323.
- Joon F. Joung. Mechanistic dataset. https://github.com/jfjoung/Mechanistic_dataset, 2025. GitHub repository, accessed 2026-07-06.

- Steven M. Kearnes, Michael R. Maser, Michael Wlekinski, Anton Kast, Abigail G. Doyle, Spencer D. Dreher, Joel M. Hawkins, Klavs F. Jensen, and Connor W. Coley. The open reaction database. *Journal of the American Chemical Society*, 143(45):18820–18826, 2021. doi: 10.1021/jacs.1c09820.
- Ross D. King, Kenneth E. Whelan, Ffion M. Jones, Philip G. K. Reiser, Christopher H. Bryant, Stephen H. Muggleton, Douglas B. Kell, and Stephen G. Oliver. Functional genomic hypothesis generation and experimentation by a robot scientist. *Nature*, 427(6971):247–252, 2004. doi: 10.1038/nature02236.
- Ryo Kuroki, Amin Mansouri, and Philippe Schwaller. Time-series as feedback: Evaluating adaptive reasoning in LLM agents. In *ICLR Workshop on Time Series in the Age of Large Models*, 2026. URL <https://openreview.net/forum?id=N3vDM8CafN>.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *arXiv preprint arXiv:2005.11401*, 2021.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *arXiv preprint arXiv:2307.03172*, 2023.
- Adrian Mirza, Nawaf Alampara, Shashank Kunchapu, Martín Ríos-García, Benjamin Emoekabu, Aravind Krishnan, Tanya Gupta, Maximilian Schilling-Wilhelmi, Michael Okereke, Anagha Aneesh, Mehrdad Asgari, Jonas Eberhardt, A. M. Elahi, H. M. Elbeheiry, M. V. Gil, C. Glaubitz, M. Greiner, C. T. Holick, T. Hoffmann, A. Ibrahim, L. C. Klepsch, Y. Köster, F. A. Kreth, J. Meyer, S. Miret, J. M. Peschel, M. Ringleb, N. C. Roesner, J. Schreiber, U. S. Schubert, L. M. Stafast, A. D. D. Wonanke, M. Pieler, Philippe Schwaller, and K. M. Jablonka. A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists. *Nature Chemistry*, 17(7):1027–1034, 2025.
- OpenAI. GPT-5 Mini model. <https://developers.openai.com/api/docs/models/gpt-5-mini>, 2025. Accessed 2026-09-24.
- Stefan Peidli, Tessa D. Green, Ciyue Shen, Torsten Gross, Joseph Min, Samuele Garda, Bo Yuan, Linus J. Schumacher, Jake P. Taylor-King, Debora S. Marks, Augustin Luna, Nils Blüthgen, and Chris Sander. scPerturb: Harmonized single-cell perturbation data. *Nature Methods*, 21(3): 531–540, 2024. doi: 10.1038/s41592-023-02144-y.
- Daniel Probst, Philippe Schwaller, and Jean-Louis Reymond. Reaction classification and yield prediction using the differential reaction fingerprint drfp. *Digital Discovery*, 1(2):91–97, 2022. doi: 10.1039/D1DD00006C.
- Qwen Team. Qwen3.5: Towards native multimodal agents. <https://qwen.ai/blog?id=qwen3.5>, 2026. Model release and model card; accessed 2026-09-24.
- Oscar Sainz, Jon Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 10776–10787, 2023. doi: 10.18653/v1/2023.findings-emnlp.722.
- Nadine Schneider, Daniel M. Lowe, Roger A. Sayle, and Gregory A. Landrum. Development of a novel fingerprint for chemical reactions and its application to large-scale reaction classification and similarity. *Journal of Chemical Information and Modeling*, 55(1):39–53, 2015. doi: 10.1021/ci5006614.
- Philippe Schwaller, Teodoro Laino, Théophile Gaudin, Peter Bolgar, Christopher A. Hunter, Costas Bekas, and Alpha A. Lee. Molecular transformer: A model for uncertainty-calibrated chemical reaction prediction. *ACS Central Science*, 5(9):1572–1583, 2019. doi: 10.1021/acscentsci.9b00576.
- Marwin H. S. Segler, Mike Preuss, and Mark P. Waller. Planning chemical syntheses with deep neural networks and symbolic AI. *Nature*, 555(7698):604–610, 2018.

- Uri Shaham, Elad Segal, Maor Ivgi, Avia Efrat, Ori Yoran, Adi Haviv, Ankit Gupta, Wenhan Xiong, Mor Geva, Jonathan Berant, and Omer Levy. SCROLLS: Standardized comparison over long language sequences. *arXiv preprint arXiv:2201.03533*, 2022.
- Rolf van der Lingen. Reaction SMILES USPTO year 2023, 2023. URL https://figshare.com/articles/dataset/Reaction_SMILES_USPTO_year_2023/24921555.
- Xingyao Wang, Yangyi Chen, Lifan Yuan, Yizhe Zhang, Yunzhu Li, Hao Peng, and Heng Ji. Executable code actions elicit better LLM agents. *arXiv preprint arXiv:2402.01030*, 2024.
- Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S. Pappu, Karl Leswing, and Vijay Pande. Moleculenet: A benchmark for molecular machine learning. *arXiv preprint arXiv:1703.00564*, 2018.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2024.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2023.
- Z.ai. GLM-5.2 model card. <https://huggingface.co/zai-org/GLM-5.2>, 2026. Accessed 2026-09-24.
- Alex L. Zhang, Tim Kraska, and Omar Khattab. Recursive language models. *arXiv preprint arXiv:2512.24601*, 2026.
- Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junying Chen, Moo Khon Hao, Xu Han, Zhen L. Thai, Shuo Wang, Zhiyuan Liu, and Maosong Sun. Infinitebench: Extending long context evaluation beyond 100k tokens. *arXiv preprint arXiv:2402.13718*, 2024.

A FULL TASK DEFINITIONS

This appendix gives the complete task definitions summarized in Table 1.

A.1 TIER 1: STRUCTURAL LOOKUP (10 QUESTIONS)

T1.1 Product Identification (10 questions): Each question names a target product by its canonical SMILES. The model must find all reaction indices in the provided context where that SMILES appears on the product side. Ten target molecules are used, each with multiple matching reactions seeded into context.

A.2 TIER 2: COMPUTED PROPERTY AGGREGATION (20 QUESTIONS)

T2.1 Molecular Weight Change (6 questions): Find all reactions where the molecular weight of the heaviest product exceeds that of the heaviest reactant by more than τ Da, for $\tau \in \{150, 184, 218, 252, 286, 320\}$. This captures reactions that substantially build up molecular mass, such as fragment couplings and macrocyclisations.

T2.2 Ring Count Change (5 questions): Find all reactions where the maximum ring-count increase across all reactant-product component pairs equals δ , for $\delta \in \{1, 2, 3, 4, 5\}$. This identifies ring-forming reactions like cyclisations, annulations, and ring-closing metatheses by degree of ring construction.

T2.3 Aromatic Ring Formation (5 questions): Find all reactions where exactly δ new aromatic rings appear in the products relative to the reactants, for $\delta \in \{1, 2, 3, 4, 5\}$. This captures aromatisation events and reactions that introduce aromatic fragments.

T2.4 Combined MW and Ring Change (4 questions): Find all reactions satisfying both a molecular weight change threshold and a ring delta simultaneously, for $(\tau, \delta) \in \{(100, 1), (100, 2), (150, 1), (150, 2)\}$. The conjunction tests the model’s ability to apply multiple filters consistently.

A.3 TIER 3: REACTION-LEVEL REASONING (35 QUESTIONS)

Bond-Level Tasks (3 questions). These tasks require tracking atomic connectivity changes between the reactant and product sides of each reaction, without needing to name the reaction class.

T3.1 Reactions that introduce at least one new nitrogen atom: product-side nitrogen count exceeds reactant-side count, as occurs when a nitrogen-containing fragment is incorporated (e.g., nitration, or amine coupling where nitrogen comes from a reagent).

T3.2 Reactions that form at least one C–N bond and break at least one C–O bond simultaneously: the atomic signature of nucleophilic substitution at carbon by nitrogen, displacing an oxygen-based leaving group (e.g., O-to-N migration or Mitsunobu-type substitutions).

T3.3 Reactions that form exactly one C–C bond with no other C–C bond changes: characteristic of selective C–C coupling reactions such as aldol additions or Grignard additions.

Scaffold-Based Tasks (3 questions). These tasks require ring-system recognition in product molecules, distinguishing reactions that construct new rings from those that merely elaborate existing ones.

T3.4 Reactions producing a product containing at least one of the fused heterocycles indole, benzofuran, benzothiazole, or benzimidazole: benzo-fused five-membered heterocycles frequent in drug scaffolds.

T3.5 Reactions that construct a new ring system in the product, identified by the appearance of an SSSR component absent from all reactant molecules.

T3.6 Reactions that build a fused bicyclic or polycyclic ring system from non-fused reactants: the product must contain two or more rings sharing at least one bond while no reactant does.

Reagent-Based Tasks (2 questions). These tasks require inspecting the middle (reagent) field of the reaction SMILES and recognizing specific reagent classes.

T3.7 Reactions whose reagent slot contains a transition metal (d-block elements, groups 3–12, excluding lanthanides and actinides), signalling metal-catalysed processes such as palladium cross-couplings, ruthenium metatheses, or copper-mediated reactions.

T3.8 Reactions whose reagent slot contains HATU+PF6 or T3P: two amide-bond activating agents widely used in peptide and medicinal chemistry synthesis.

Stereochemistry-Based Tasks (2 questions). Correct interpretation requires understanding CIP priority rules and reconstructing three-dimensional structure from the SMILES string.

T3.9 Reactions where the product contains at least one tetrahedral stereocenter (explicit R/S annotation) and none of the reactants contain any: reactions that create stereocenters de novo, such as asymmetric reductions or chiral auxiliary-controlled additions.

T3.10 Reactions producing a product with an E-configured (trans) C=C double bond (explicit CIP annotation, non-aromatic): typically eliminations, Wittig-type olefinations, or geometrically selective cross-metatheses.

Functional-Group Tasks (15 questions). These tasks use SMIRKS reaction templates from the Rxn-INSIGHT library (Dobbelaere et al., 2024), so identification requires verifying that the correct bond-forming and bond-breaking events occurred in the right atomic context.

T3.11–T3.14 Amide coupling variants (4 questions): Four mechanistically distinct routes to an amide bond: acyl chloride with primary amine (Schotten–Baumann); carboxylic acid with primary amine (condensation); ester aminolysis with primary amine; ester aminolysis with secondary amine.

T3.15–T3.19 Functional-group transformations (5 questions): Grignard addition to ketone giving tertiary alcohol; nitrile reduction to primary amine; nitro reduction to amine; alcohol conversion to organic azide; primary alcohol oxidation to carboxylic acid.

T3.20–T3.25 Protecting-group and named-reaction transformations (6 questions): Boc deprotection of a secondary amine; Boc protection of a primary amine (2 questions); Suzuki coupling with a boronic acid; Mitsunobu reaction on a sulfonamide nucleophile; Buchwald–Hartwig / Ullmann–Goldberg N-arylation of a primary amine; Stille coupling with an aryl stannane (4 questions).

Reaction Mechanism Tasks (10 questions). These tasks use mechanism-level templates from a mechanistic reaction dataset (Joung, 2025), requiring that the full bond-making and bond-breaking sequence of a named mechanistic pathway be matched.

T3.26–T3.35 Named mechanistic pathways (10 questions): Ester hydrolysis via hydroxide-promoted acyl substitution; Mitsunobu reaction; Wittig olefination; Knoevenagel condensation / aldol dehydration; azide-alkyne Huisgen [3+2] cycloaddition; keto alpha-alkylation (enolate C-alkylation); base-catalysed transesterification; Grignard carbonyl addition; Staudinger reduction; alpha-ketone electrophilic bromination.

A.4 TIER 4: PAIRWISE AND RELATIONAL REASONING (35 QUESTIONS)

Throughout, chains are ordered reaction sequences where a product of step i equals a reactant of step $i + 1$ under exact canonical-SMILES equality.

Mechanical Graph Tasks (5 questions).

T4.1–T4.2 Synthetic chain enumeration (2 questions): Find all ordered synthesis chains of fixed length (3 and 4 reactions) starting from a given reaction index.

T4.3–T4.4 Longest synthesis chain (2 questions): Given a target product SMILES, find the longest chain of reactions with ascending indices that terminates in that product.

T4.5 Hub molecule identification (1 question): Find all molecules that appear as the product of exactly one reaction and as a reactant in at least three distinct downstream reactions (higher indices), analogous to branch-point compounds in multi-target synthesis campaigns.

Chemically Constrained Chain Tasks (10 questions).

T4.6–T4.9 Functional-group transformation chains (4 questions): Find all 7-reaction chains where the entry reaction carries a specified source functional group among its reactants and the exit reaction produces a specified target functional group. The four source-to-target pairs are: primary alcohol $\rightarrow$ carboxylic acid; alkyl halide $\rightarrow$ tertiary amine; ester $\rightarrow$ tertiary amide; nitrile $\rightarrow$ primary amide.

T4.10–T4.11 Protecting-group pairs (2 questions): Find all reaction pairs ($r_{\text{install}}, r_{\text{remove}}$) with $r_{\text{install}} < r_{\text{remove}}$ where the same protecting group is installed on a functional group in one reaction and removed in a later reaction on the same canonical molecular scaffold. Two protecting groups are queried: Boc on nitrogen; benzyl on oxygen/nitrogen.

T4.12–T4.15 Ring-construction chains (4 questions): Find all reaction chains of a fixed length where an acyclic precursor is progressively elaborated into a product containing a specified fused ring system: quinoline, indole, benzothiazole, or benzimidazole.

Prospective and Multi-Constraint Reasoning Tasks (20 questions). Both task groups require interacting constraints across multiple reactions and therefore extend beyond a single linear scan with one filter per reaction.

T4.16–T4.25 Truncated synthesis routes (10 questions): A ground-truth 5-reaction synthesis route leading to a named target molecule is mined from the full dataset. At evaluation time, only the first four reactions of each route appear in context; the final step producing the target is withheld. The model must find all valid 4-reaction prefixes of complete 5-step routes to that target. Because the target never appears as a product in context, the model must identify prefixes compatible with an unseen final step. The prompt supplies only the target name, so performance also depends on name-to-structure interpretation. Ten target molecules span heterocyclic, biaryl, peptidic, spirocyclic, and lactam scaffolds.

T4.26–T4.35 SMIRKS-constrained 2-step synthesis chains (10 questions): Find all ordered 2-reaction chains $[r_0, r_1]$ where step 1 matches a specified SMIRKS template, step 2 matches a second template, the product of r_0 and a reactant of r_1 share the same canonical SMILES, and an additional chemical-persistence condition is satisfied. The 10 reaction-type pairs include Boc protection $\rightarrow$ Buchwald–Hartwig, nitro reduction $\rightarrow$ amide coupling, Suzuki coupling $\rightarrow$ carbonyl reduction, Wittig olefination $\rightarrow$ alkene hydrogenation, alcohol oxidation $\rightarrow$ reductive amination, azide formation $\rightarrow$ Staudinger reduction, and bromination/boronation $\rightarrow$ coupling sequences.

A.5 FULL-CORPUS ANSWER-SET CARDINALITY

Table 4 reports the number of elements in the ground-truth set scored by F1 at full scale. For the longest-chain task, the scored elements are reaction indices in the returned chain; for the other Tier 4 tasks they are complete chains, reaction pairs, or molecules. The prospective subgroup has larger sets than the mechanical subgroup, so cardinality could contribute to that pairwise contrast. It nevertheless has only 3.1% as many answers on average as the chemically constrained subgroup, despite much lower F1 (0.045 versus 0.459), ruling out answer-set cardinality as a sufficient explanation of the subgroup ordering.

Table 4: Full-corpus Tier 4 ground-truth cardinality, computed from the stored task constants. Statistics are across questions.

Subgroup	Q	Mean $ G $	Median $ G $	Range
Mechanical graph	5	8.8	8.0	8–10
Chemically constrained	10	979.6	191.5	1–5,378
Prospective / multi-constraint	20	30.7	12.0	2–190

A.6 MATCHED-CARDINALITY DIAGNOSTICS

The completed GPT-5 mini control contains 725 jobs and 2,175 question-level trajectories. Figure 4 exposes the task-level heterogeneity behind the tier means in Figure 3a–b. The fixed- K degradation is concentrated in chemistry-intensive Tier-3 families, while deterministic Tier-2 filters remain close to perfect. Figure 5 reports the corresponding calls, input and output tokens, billed cost, latency, tool time, wall time, and peak memory.

A.7 CHEMISTRY-RULE CONTROL BY TASK

Table 5 disaggregates the full-corpus control in Figure 3c–d. Supplying the executable chemistry rule strongly improves reaction-class and mechanism recognition for both models. The effect varies by operation: Qwen3.5-397B-A17B’s stereochemistry score falls from 1.000 to 0.000, and the graph-task effects are mixed. Because deterministic exhaustive execution of each supplied rule scores 1.000, these residual errors cannot be attributed to incorrect rules or scoring. They show that an agent may fail to apply even a correct local rule consistently across the corpus or within a larger graph-search procedure.

Figure 7 gives the complete task-by-context breakdown. Figures 8 and 9 report the associated resource profiles. Supplying the local rule reduces calls and tokens for Qwen3.5-397B-A17B at the two sampled contexts, but the full-corpus graph tasks remain expensive and the performance effect depends on the task. Claude Haiku 4.5 shows the same task dependence, with the largest full-corpus improvements on reaction-class and mechanism recognition.

A.7.1 DETERMINISTIC EXECUTOR CEILING.

All 15 executor jobs—five task configurations at three context sizes—reach macro-F1 and exact-set accuracy of 1.000 with zero model calls, tokens, and API cost. Figure 10 shows that deterministic runtime grows at full-corpus scale, particularly for the two graph operations, while peak memory

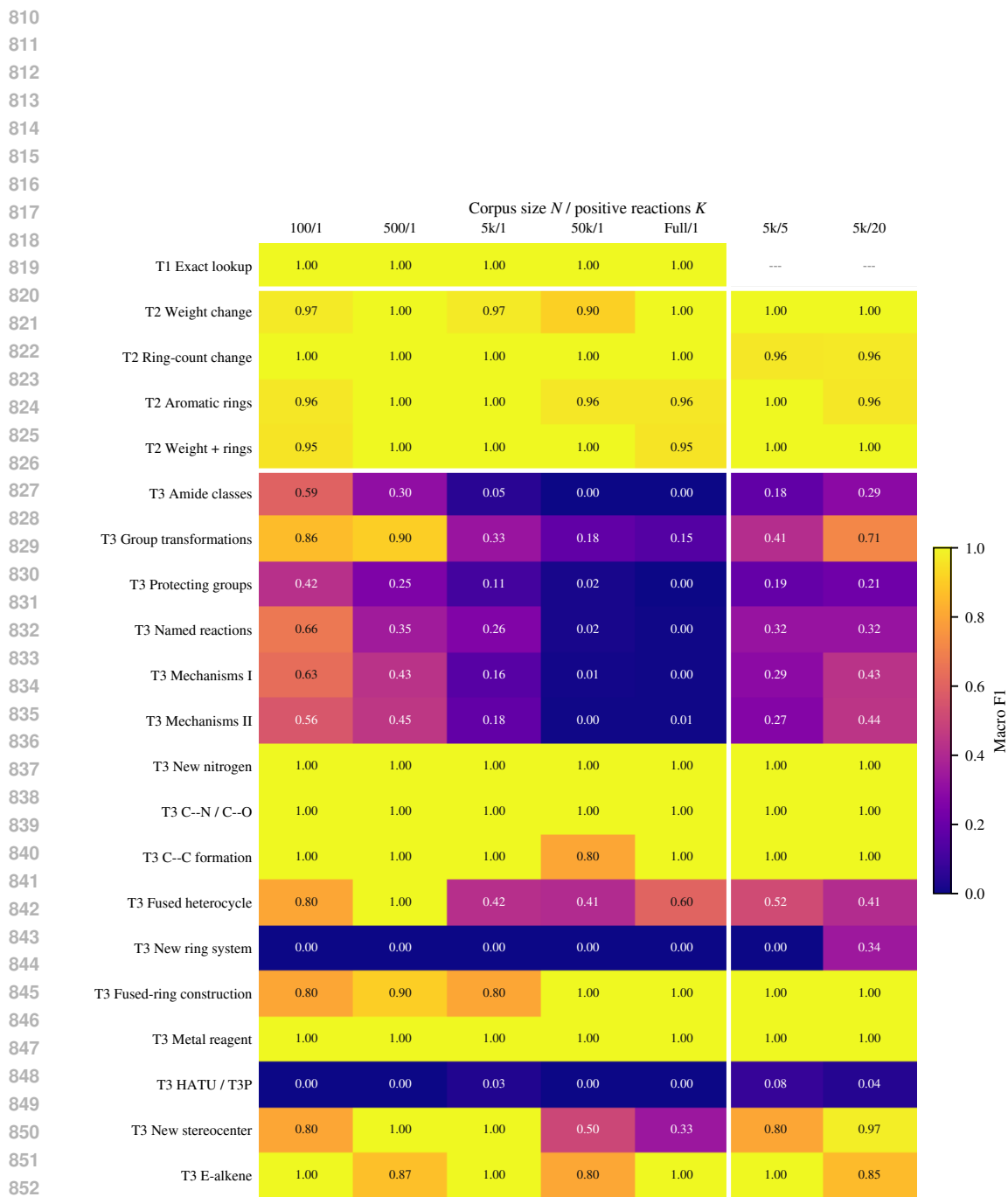


Figure 4: GPT-5 mini matched-cardinality macro-F1 by task. The first five columns increase corpus size N with one positive reaction ($K = 1$); the final two hold $N = 5,000$ and increase K . Each cell averages five repetitions. Tier 1 was evaluated only in the fixed- K scale sweep.

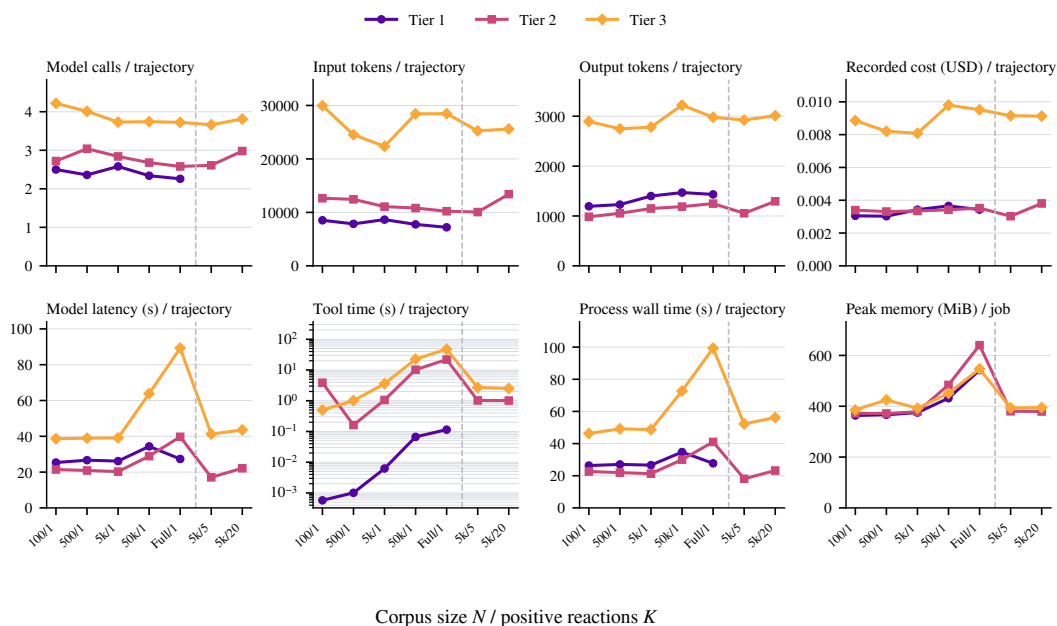


Figure 5: GPT-5 mini matched-cardinality resource profile. Additive quantities are normalized per successfully scored question trajectory; peak combined host-process-tree plus Docker memory is averaged per job. The dashed divider separates the fixed- K scale sweep from the fixed- N cardinality sweep. Input and output tokens are reported separately.

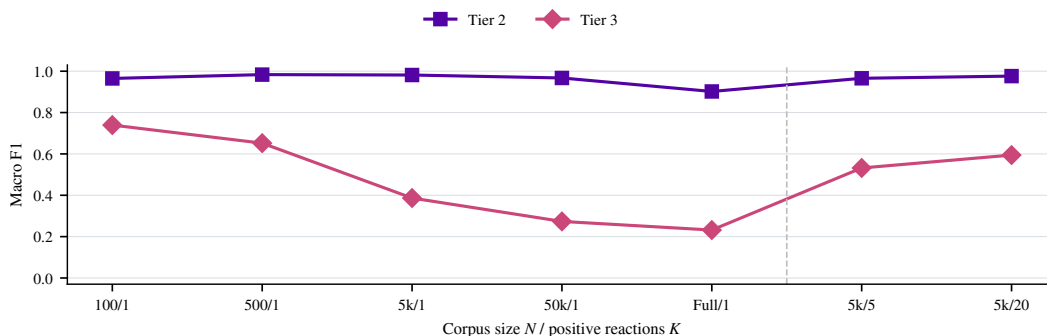


Figure 6: Qwen3.5-397B-A17B matched-cardinality macro-F1 over completed successful runs. Condition labels give corpus size N and positive cardinality K as N/K ; the vertical divider separates corpus-scaling conditions from fixed-corpus cardinality conditions.

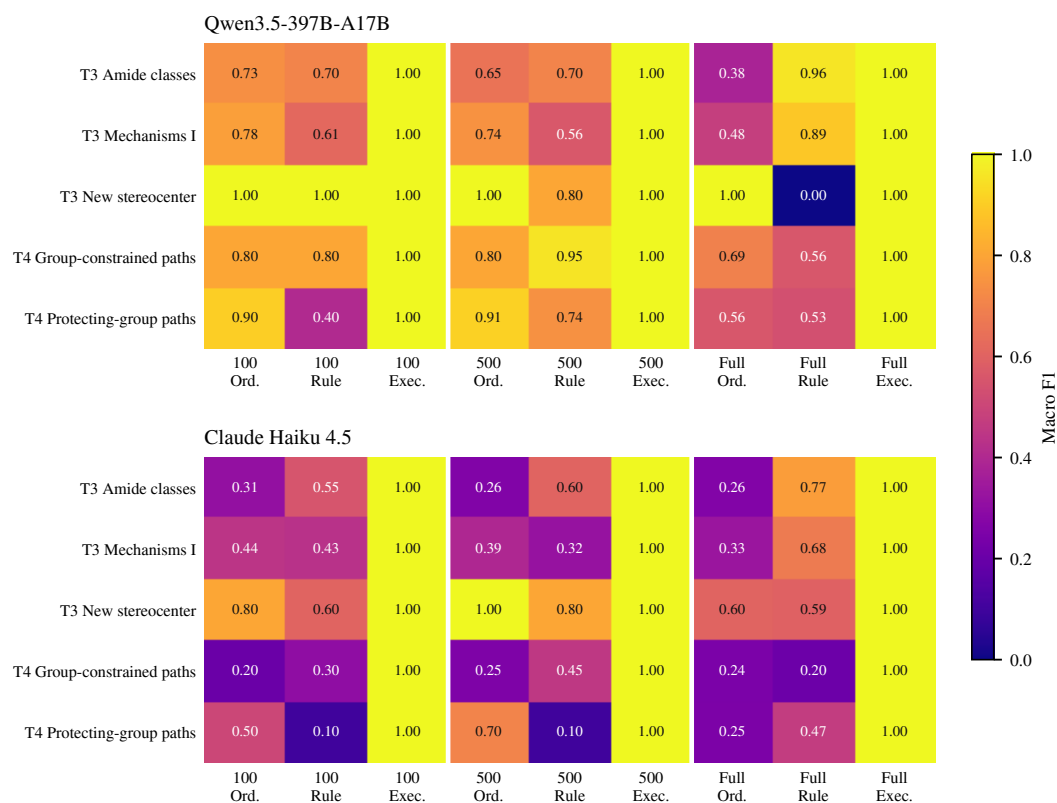


Figure 7: Chemistry-rule control by task, context, and model. Ordinary RLM (Ord.) and rule-supplied RLM (Rule) are averaged over five repetitions. Deterministic exhaustive execution (Exec.) applies the same rule to every record and provides the verified ceiling.

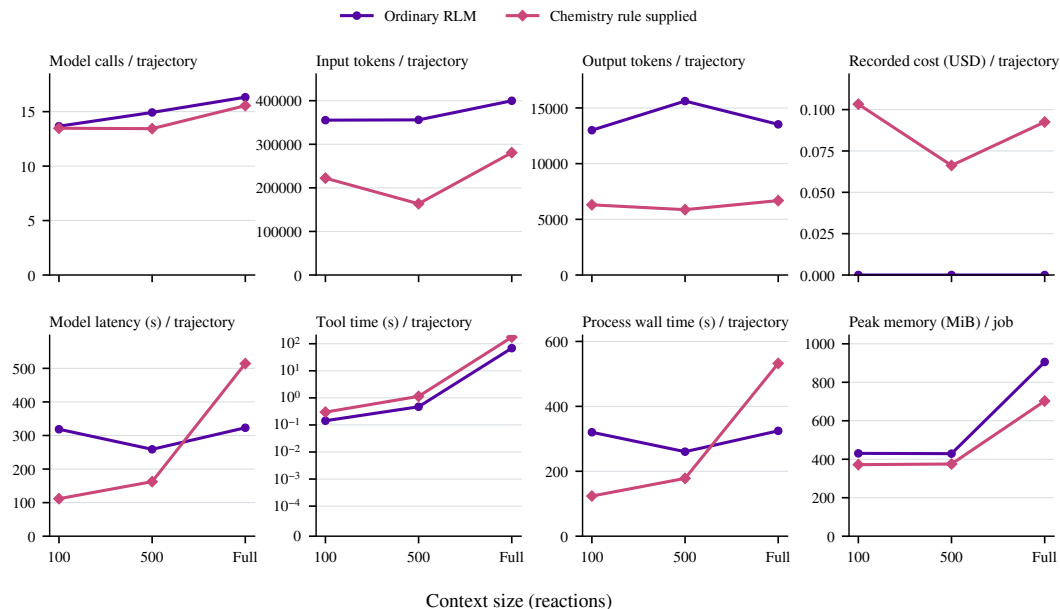


Figure 8: Qwen3.5-397B-A17B chemistry-rule resource profile. Calls, tokens, billed cost, recorded model latency, tool time, and process wall time are normalized per successfully scored question trajectory; peak memory is averaged per job. Most Qwen3.5-397B-A17B runs used institutional inference and record zero API cost.

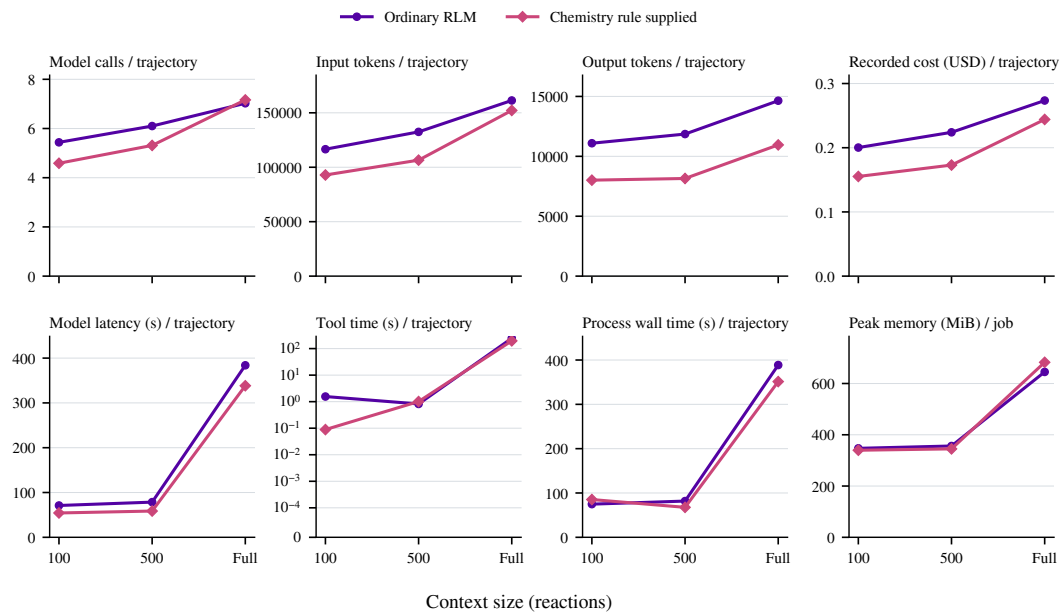


Figure 9: Claude Haiku 4.5 chemistry-rule resource profile, with the same aggregation and normalization as Figure 8.

Table 5: Full-corpus macro-F1 for ordinary RLM (Ord.), RLM supplied with the executable chemistry rule (Rule), and deterministic exhaustive execution (Exec.). Values are question-weighted means over five runs. Q is the number of questions in each task configuration.

Task	Q	Qwen3.5-397B-A17B		Claude Haiku 4.5		Exec.
		Ord.	Rule	Ord.	Rule	
T3 reaction class (Task 6)	4	0.413	1.000	0.297	0.800	1.000
T3 mechanisms (Task 10)	5	0.458	0.960	0.332	0.720	1.000
T3 stereochemistry (Task 23)	1	1.000	0.000	0.599	0.603	1.000
T4 functional-group paths (Task 13)	4	0.692	0.556	0.243	0.204	1.000
T4 protecting-group paths (Task 14)	2	0.556	0.533	0.247	0.467	1.000

remains below 341 MiB. This verifies the supplied rules, exhaustive execution path, and scorers independently of model behavior.

A.8 TASK-16 DECOMPOSITION BY MODEL AND TARGET

Figure 11 complements Table 3 with precision, recall, and exact-route accuracy. Figure 12 reveals that the aggregate gain from the final-step class is target-dependent: Qwen3.5-397B-A17B improves on the lactam dipeptide and benzamide-pyrazole targets, while Claude Haiku 4.5’s gain is concentrated on benzamide-pyrazole. Neither model recovers the pyrimidine-piperazine route from the structure-plus-class prompt. Figure 13 reports the computational profile of all 30 jobs and 90 target-level trajectories.

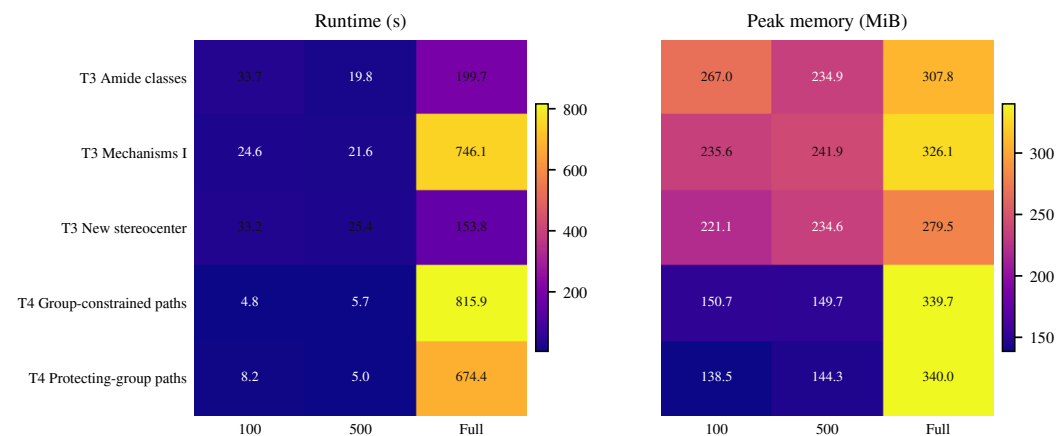


Figure 10: Runtime and peak memory for deterministic exhaustive execution of the five chemist-validated rules. Every cell scores macro-F1 and exact-set accuracy of 1.000 and uses no language-model calls.

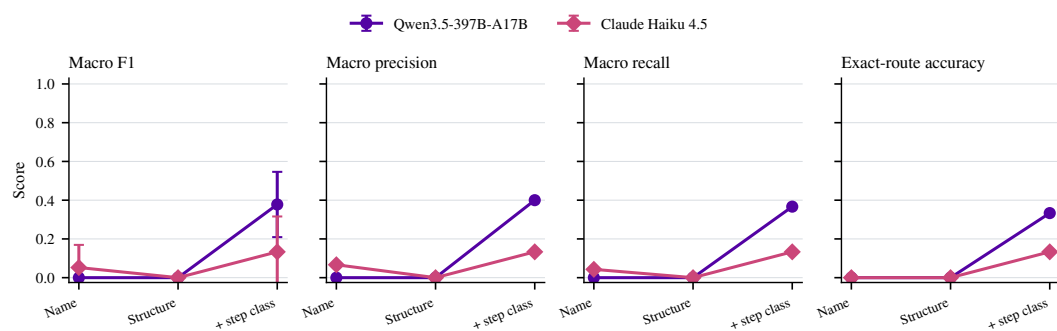


Figure 11: Aggregate Task-16 decomposition across three targets and five repetitions. Error bars on macro-F1 show one run-level standard deviation. Exact-route accuracy requires the complete predicted chain set to equal the reference set.

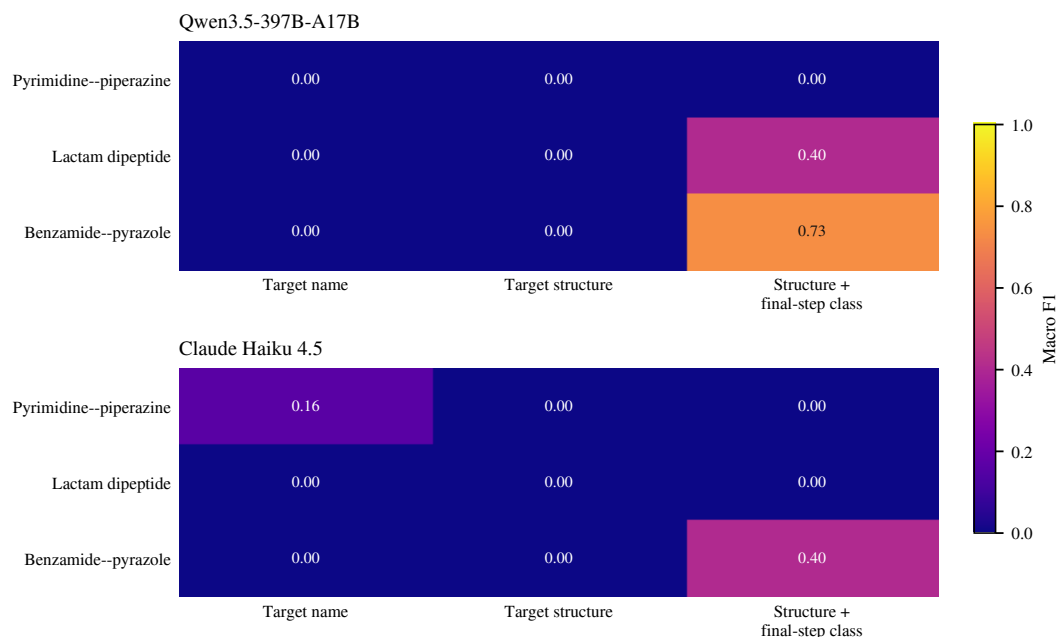


Figure 12: Task-16 macro-F1 separated by model, target, and prompt condition. Each cell averages five repetitions after removing every reaction that directly produces the target.

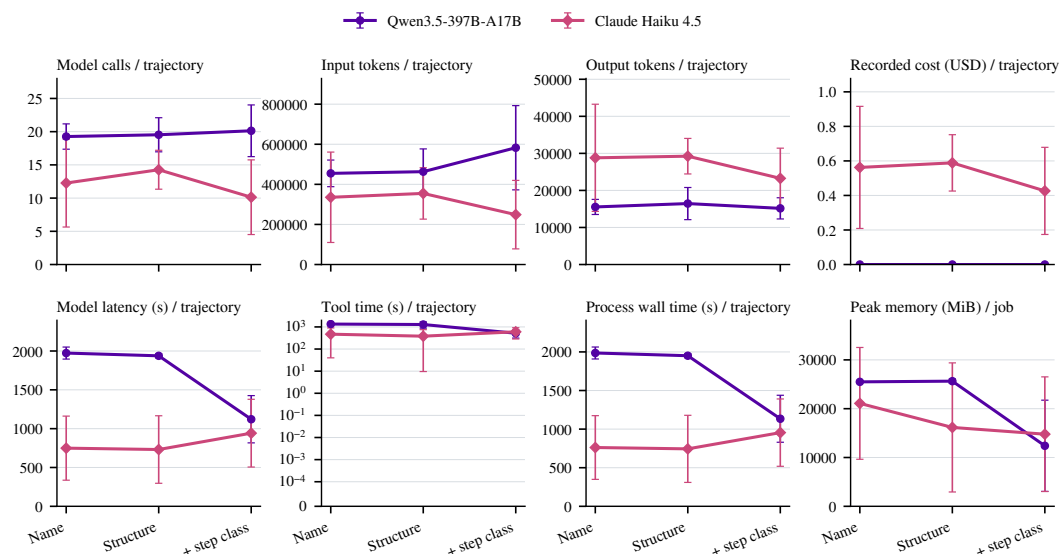


Figure 13: Task-16 resource diagnostics. Calls, tokens, billed cost, model latency, tool time, and wall time are normalized per target-level trajectory; peak combined memory is averaged per job. Points show means over five jobs and error bars show one standard deviation.

B PERFORMANCE AND RESOURCE PROFILES

Figure 14 shows the accuracy–cost trade-off across Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5, the three models with complete paid-API accounting for the core benchmark. Figure 15 separates the same observations by model. The profiles that follow report performance, cost, tokens, time, memory, and task-level variation from the frozen per-run records.

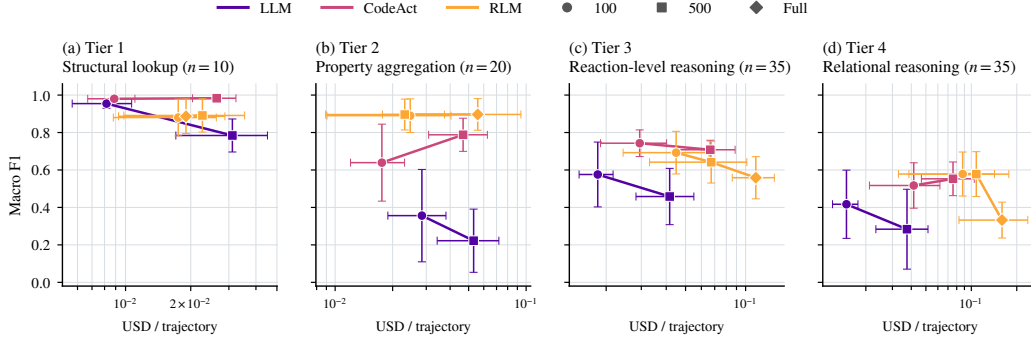


Figure 14: Cross-model accuracy–cost trade-off by tier for Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5. Points are unweighted means across the three model-level values; horizontal and vertical bars show standard errors across models. Color identifies the access interface, marker shape identifies context size, and lines connect context settings within an interface. Cost is recorded in USD per question trajectory on a logarithmic axis.

B.1 EFFICIENCY BEYOND DOLLAR COST

Figure 16 characterizes RLM efficiency using 2,246 successful jobs, comprising 7,460 question-level trajectories from Qwen3.5-397B-A17B, DeepSeek-V4-Flash-0731, Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5. For each model, calls, tokens, recorded latency, tool time, and process wall time are normalized per successful trajectory, while peak memory is averaged per successful job. The plotted points are unweighted means across the five models, with standard errors capturing model-to-model variation. At Tier 4, moving RLM from $x = 500$ to full-corpus access raises the across-model mean from 11.79 to 13.68 calls, 354k to 411k tokens, 238 to 704 seconds of recorded latency, 8.22 to 319 seconds of tool execution, 241 to 708 seconds of process wall time per successful trajectory, and 413 MiB to 3.50 GiB peak memory per job. The latency, tool-time, and process-wall counters are independently recorded and represent separate measurements. Every trajectory in this analysis has a complete record for all six measures.

B.2 QWEN3.5-397B-A17B PROFILE

Figures 17–19 report the Qwen3.5-397B-A17B benchmark profile. The main evaluation contains 1,050 terminal jobs and 3,500 question-level trajectories: plain LLM and CodeAct at $x \in \{100, 500\}$, and RLM at $x \in \{100, 500, \text{full}\}$. The extended-scale CodeAct evaluation adds 150 successful jobs and 500 trajectories at $x = 1000$. One main-matrix RLM job ended without a valid answer and contributes zero to macro-F1. Additive resource measures use successfully scored question trajectories, and peak memory is averaged over successful jobs.

B.3 COMPLETE DEEPSEEK-V4-FLASH-0731 PROFILE

Figures 20–22 report the complete DeepSeek-V4-Flash-0731 benchmark profile. The main evaluation contains 1,050 terminal jobs and 3,500 question-level trajectories: plain LLM and CodeAct at $x \in \{100, 500\}$, and RLM at $x \in \{100, 500, \text{full}\}$. The extended-scale evaluation adds CodeAct and RLM at $x = 1000$, with 150 jobs and 500 trajectories per interface. The standard and extended CodeAct arms contain 31 and 65 terminal failures; these contribute zero to macro-F1. All 450

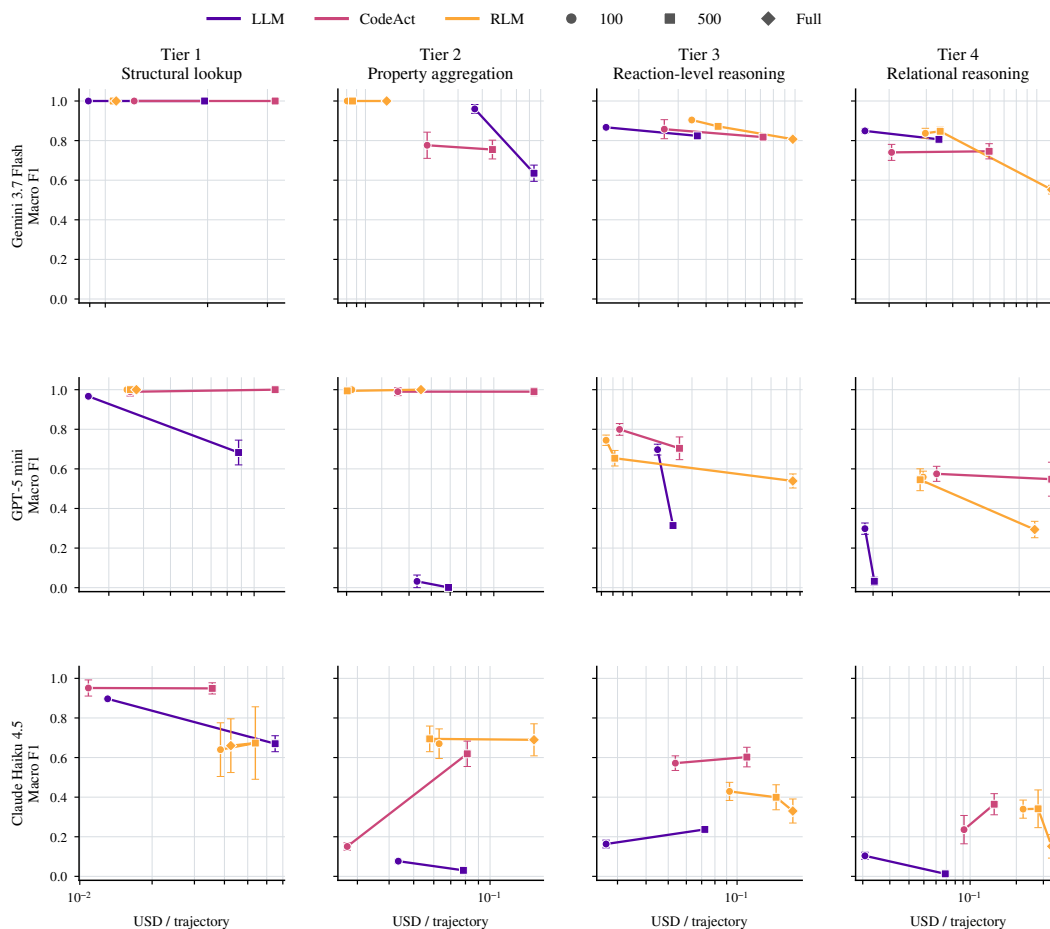


Figure 15: Model-separated accuracy–cost frontiers for Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5. Each row is one model and each column one benchmark tier. Points show means over five repetitions, vertical bars show one population standard deviation, color identifies the access interface, and marker shape identifies context size.

1296
1297
1298
1299
1300
1301
1302
1303
1304
1305
1306
1307
1308
1309
1310
1311
1312
1313
1314
1315
1316
1317
1318
1319
1320
1321
1322
1323
1324
1325
1326
1327
1328
1329
1330
1331
1332
1333
1334
1335
1336
1337
1338
1339
1340
1341
1342
1343
1344
1345
1346
1347
1348
1349

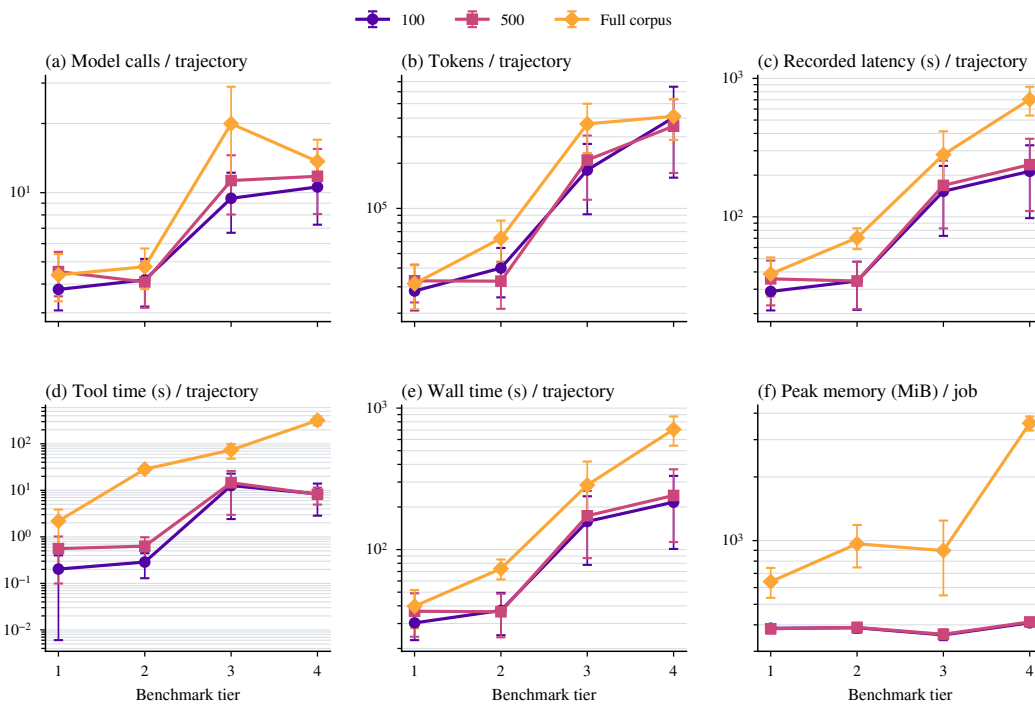


Figure 16: RLM efficiency beyond billed cost. Curves show unweighted means and standard errors across Qwen3.5-397B-A17B, DeepSeek-V4-Flash-0731, Gemini 3.7 Flash, GPT-5 mini, and Claude Haiku 4.5. Calls, tokens, recorded latency, tool time, and wall time are normalized per successful scored question trajectory; peak combined host-process-tree plus Docker memory is averaged per successful job.

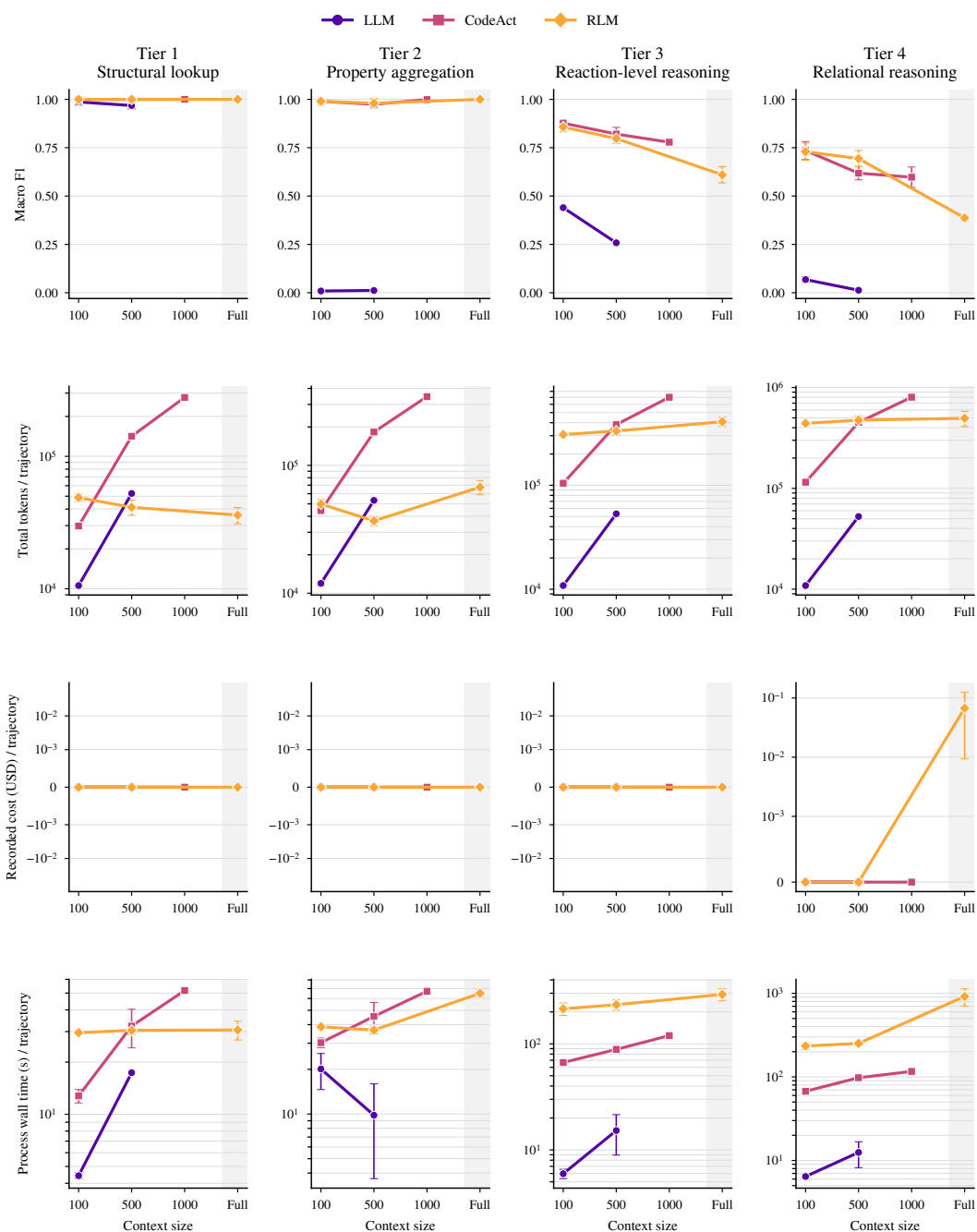


Figure 17: Qwen3.5-397B-A17B performance and efficiency by tier. Each row contains four tier panels. Bars show one population standard deviation across five repetitions. Token and process-wall-time means are normalized per successfully scored question trajectory. Most runs used institutional inference and record no API charge; five full-corpus RLM jobs used a paid endpoint and retain their billed cost in USD. The $x = 1000$ extension covers CodeAct, while full-corpus access is RLM-only.

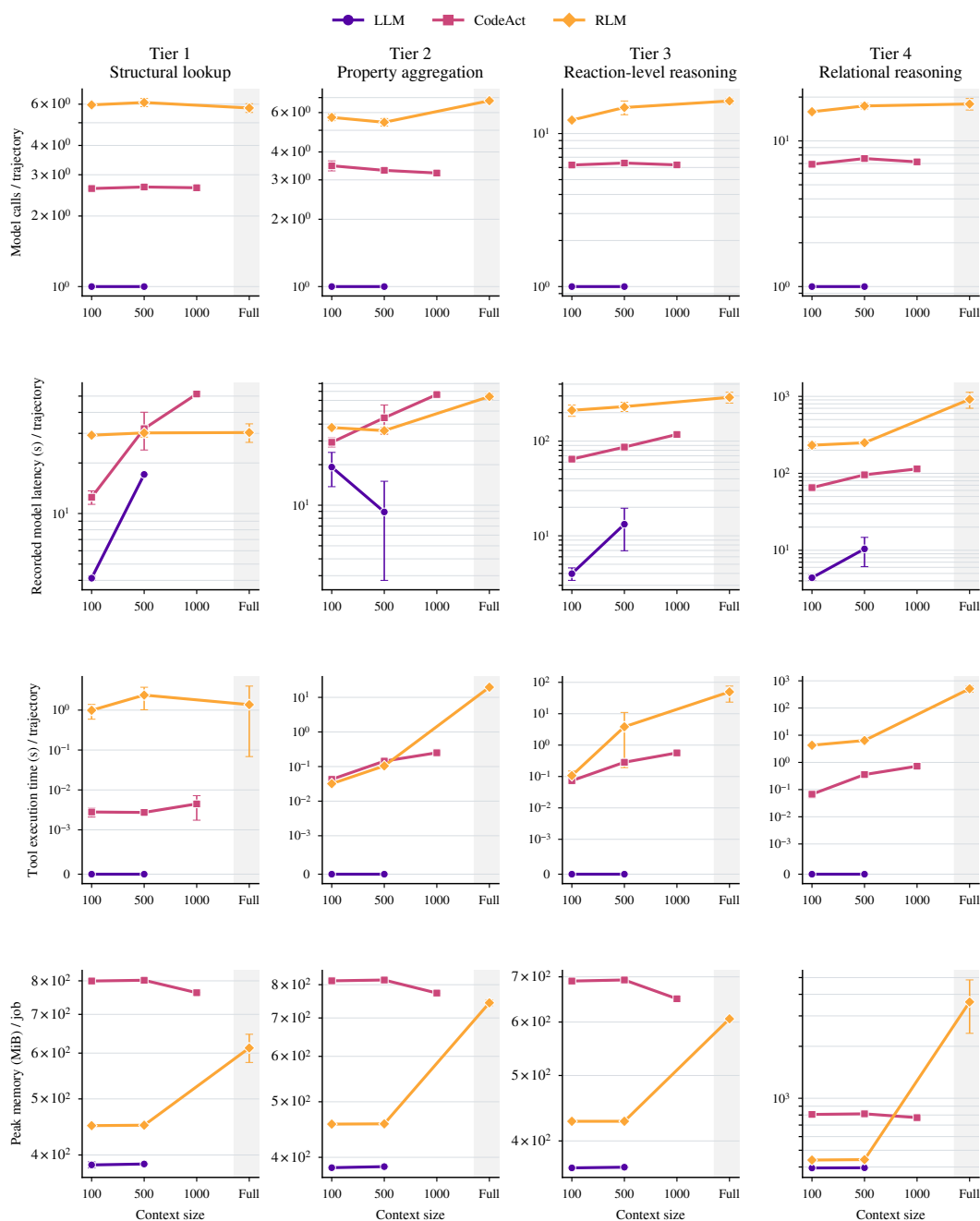


Figure 18: Qwen3.5-397B-A17B resource diagnostics by tier. Model calls count successful language-model API calls, recorded latency accumulates model-call latency, and tool time records local Python/RDKit execution. These additive measures are normalized per successfully scored question trajectory. Peak memory is the mean per-job maximum of the combined host process tree and Docker processes.

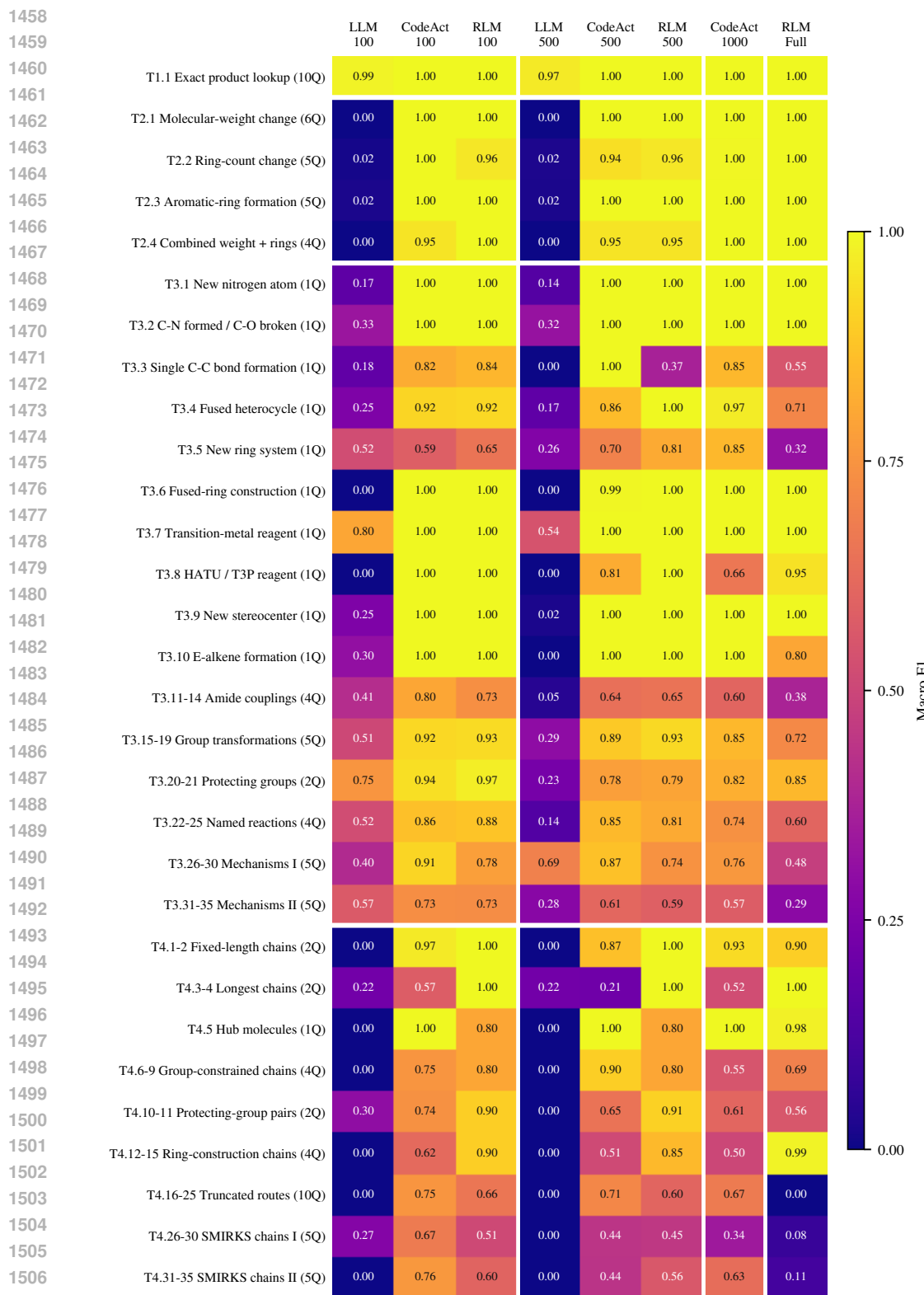


Figure 19: Qwen3.5-397B-A17B task-level macro-F1, averaged across five repetitions. The terminal failure contributes zero. The heatmap preserves all 30 task configurations and includes the $x = 1000$ CodeAct extension; full-corpus access is RLM-only.

standard RLM jobs and all 150 RLM $x = 1000$ jobs succeeded. Additive resource measures use successfully scored question trajectories, and peak memory is averaged over successful jobs.

B.4 COMPLETE CLAUDE HAIKU 4.5 PROFILE

Figures 23–25 report the complete Claude Haiku 4.5 benchmark profile. The evaluation contains 1,050 successful jobs and 3,500 question-level trajectories: plain LLM and CodeAct at $x \in \{100, 500\}$, and RLM at $x \in \{100, 500, \text{full}\}$. Each point in the tier summaries averages five repetitions. Macro-F1 is question-weighted within each repetition; additive resource measures are normalized by the number of successfully scored question trajectories, and peak memory is averaged over successful jobs.

B.5 COMPLETE GEMINI 3.7 FLASH PROFILE

Figures 26–28 report the complete Gemini 3.7 Flash benchmark profile. The main evaluation contains 1,050 terminal jobs and 3,500 question-level trajectories: plain LLM and CodeAct at $x \in \{100, 500\}$, and RLM at $x \in \{100, 500, \text{full}\}$. The extended-scale evaluation adds CodeAct and RLM at $x = 1000$, with 150 jobs and 500 trajectories per interface. Terminal failed jobs contribute zero to macro-F1. Additive resource measures use successfully scored question trajectories, and peak memory is averaged over successful jobs.

B.6 GPT-5 MINI PROFILE

Figures 29–31 report the GPT-5 mini benchmark profile. The main evaluation contains 1,050 successful jobs and 3,500 question-level trajectories: plain LLM and CodeAct at $x \in \{100, 500\}$, and RLM at $x \in \{100, 500, \text{full}\}$. Extended-scale CodeAct and RLM evaluations each add 150 jobs and 500 trajectories at $x = 1000$. Terminal failed jobs contribute zero to macro-F1. Additive resource measures use successfully scored question trajectories, and peak memory is averaged over successful jobs.

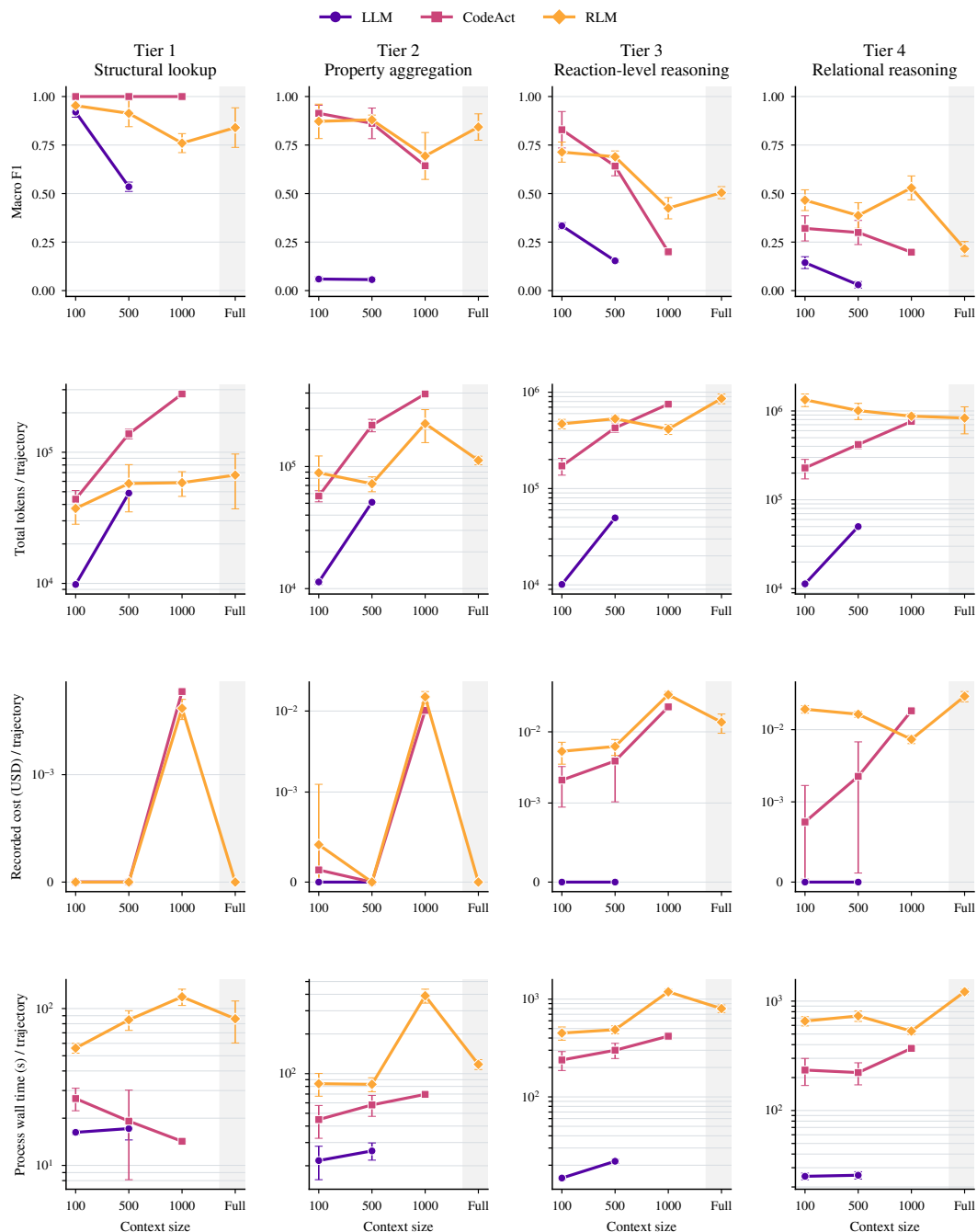


Figure 20: DeepSeek-V4-Flash-0731 performance and efficiency by tier. Each row contains four tier panels. Bars show one population standard deviation across five repetitions. Recorded cost is reported in USD; institutional-inference runs record no API charge, while paid-endpoint continuations retain their billed cost. Token and process-wall-time means are normalized per successfully scored question trajectory. The $x = 1000$ extension covers CodeAct and RLM, while full-corpus access is RLM-only.

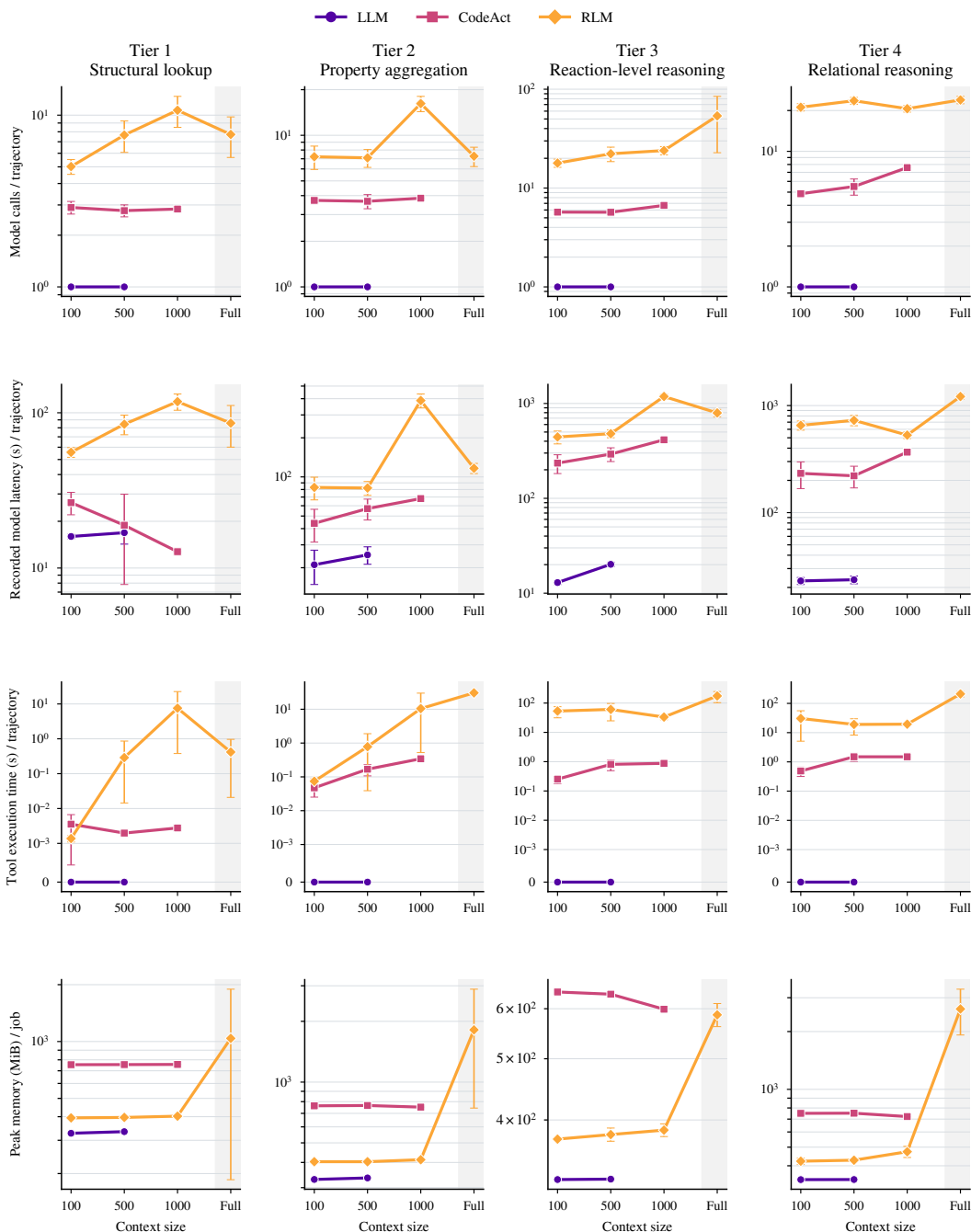


Figure 21: DeepSeek-V4-Flash-0731 resource diagnostics by tier. Model calls count successful language-model API calls, recorded latency accumulates model-call latency, and tool time records local Python/RDKit execution. These additive measures are normalized per successfully scored question trajectory. Peak memory is the mean per-job maximum of the combined host process tree and Docker processes.

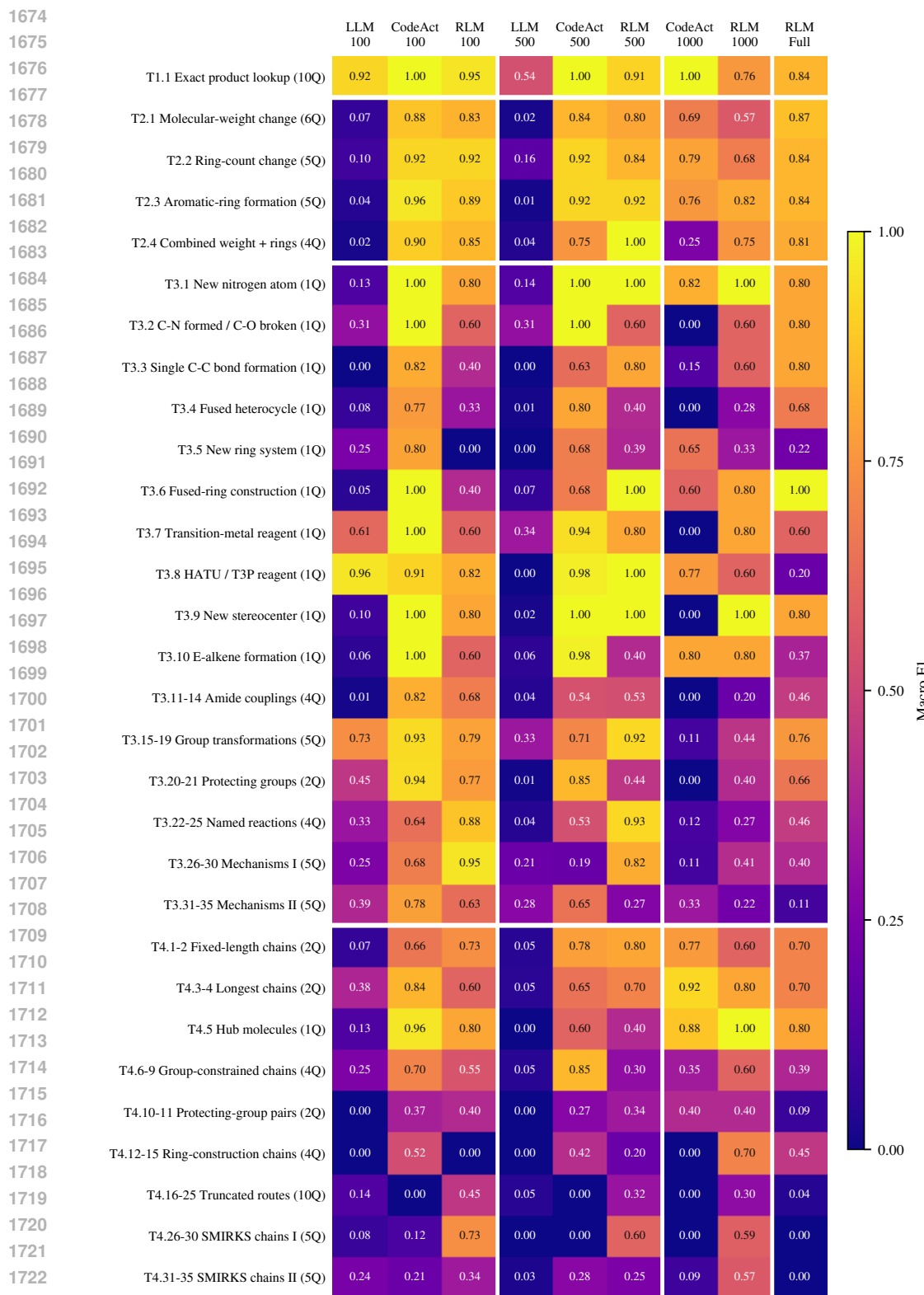


Figure 22: DeepSeek-V4-Flash-0731 task-level macro-F1, averaged across five repetitions. Terminal failures contribute zero. The heatmap preserves all 30 task configurations and includes the $x = 1000$ CodeAct and RLM extensions; full-corpus access is RLM-only.

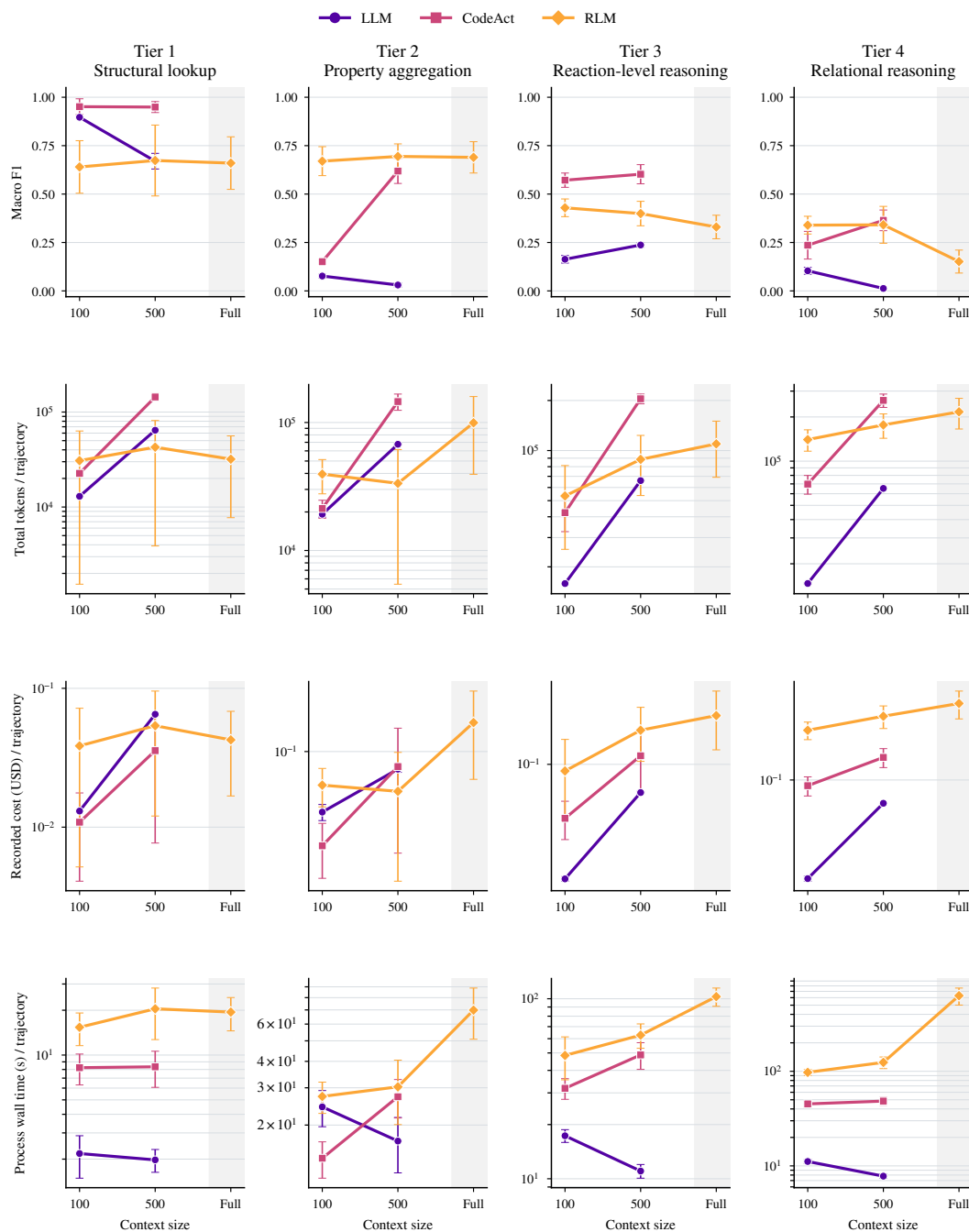


Figure 23: Claude Haiku 4.5 performance and efficiency by tier. Each row contains four tier panels. Bars show one population standard deviation across five repetitions. Cost is recorded billed cost in USD; token and process-wall-time means are normalized per successfully scored question trajectory. Full-corpus access is evaluated only for RLM.

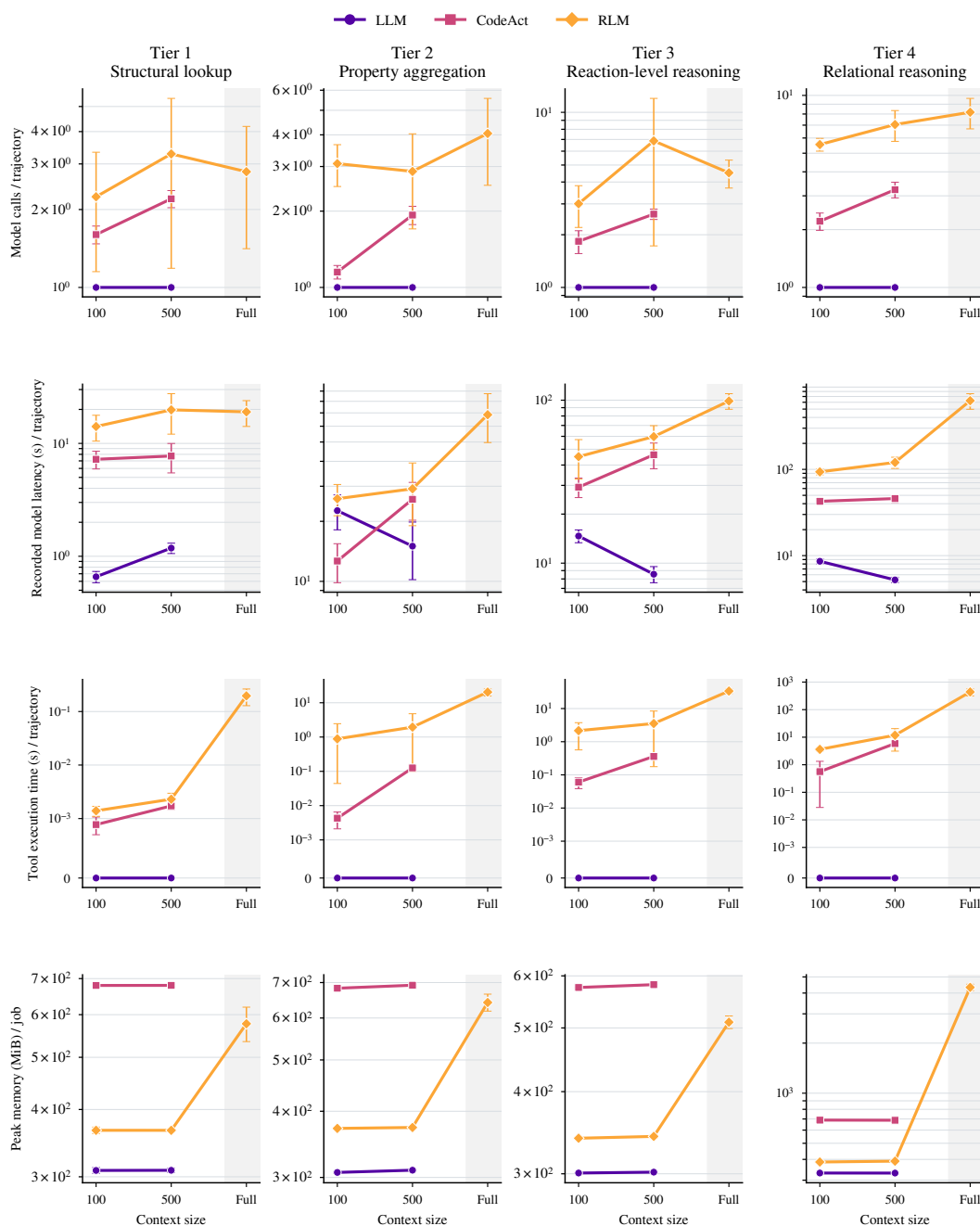


Figure 24: Claude Haiku 4.5 resource diagnostics by tier. Model calls count successful language-model API calls, recorded latency accumulates model-call latency, and tool time records local Python/RDKit execution. These additive measures are normalized per successfully scored question trajectory. Peak memory is the mean per-job maximum of the combined host process tree and Docker processes.

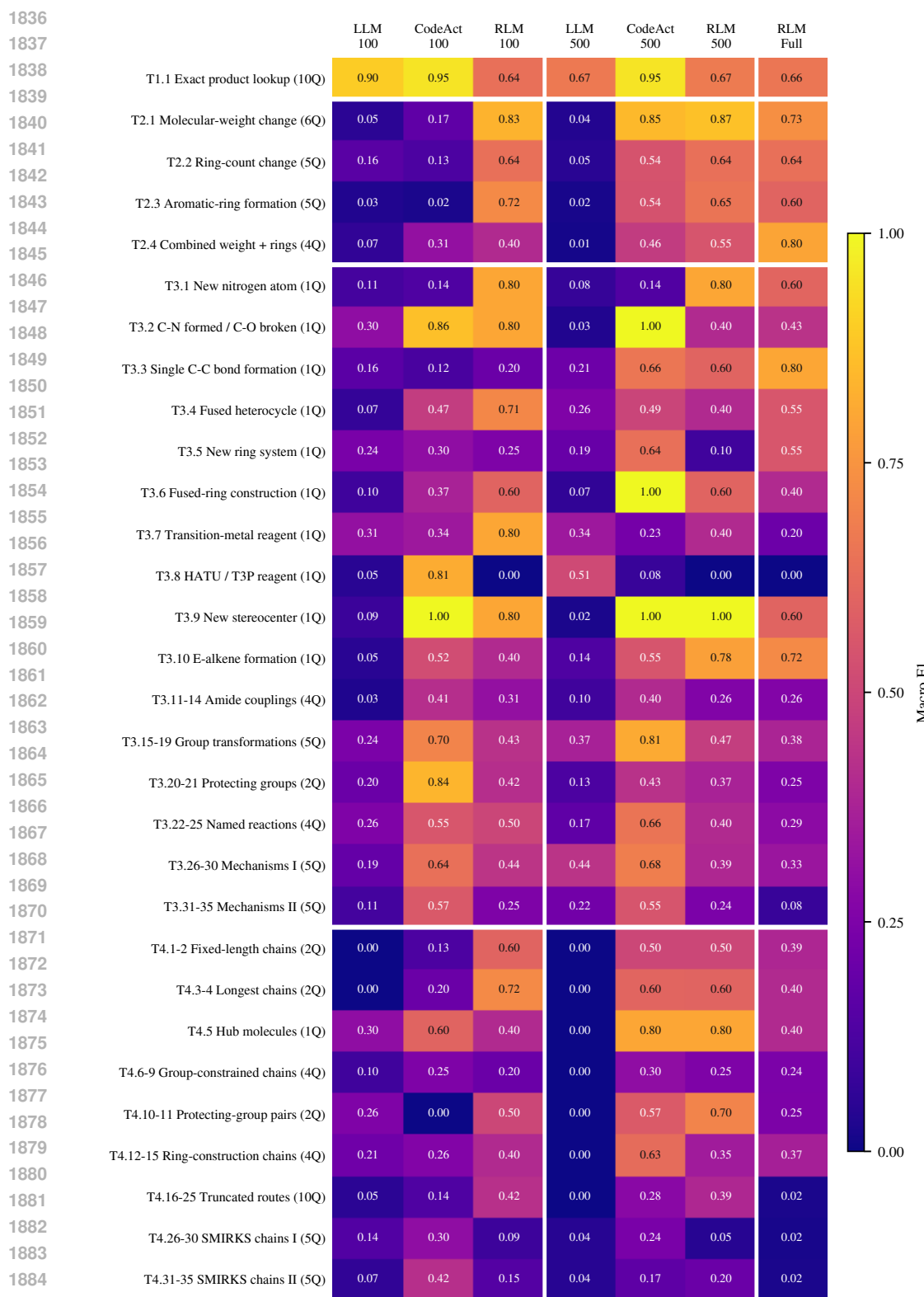


Figure 25: Claude Haiku 4.5 task-level macro-F1, averaged across five repetitions. The heatmap preserves the individual task configurations hidden by tier-level aggregation. Each column is one evaluated interface and context setting; full-corpus access is RLM-only.

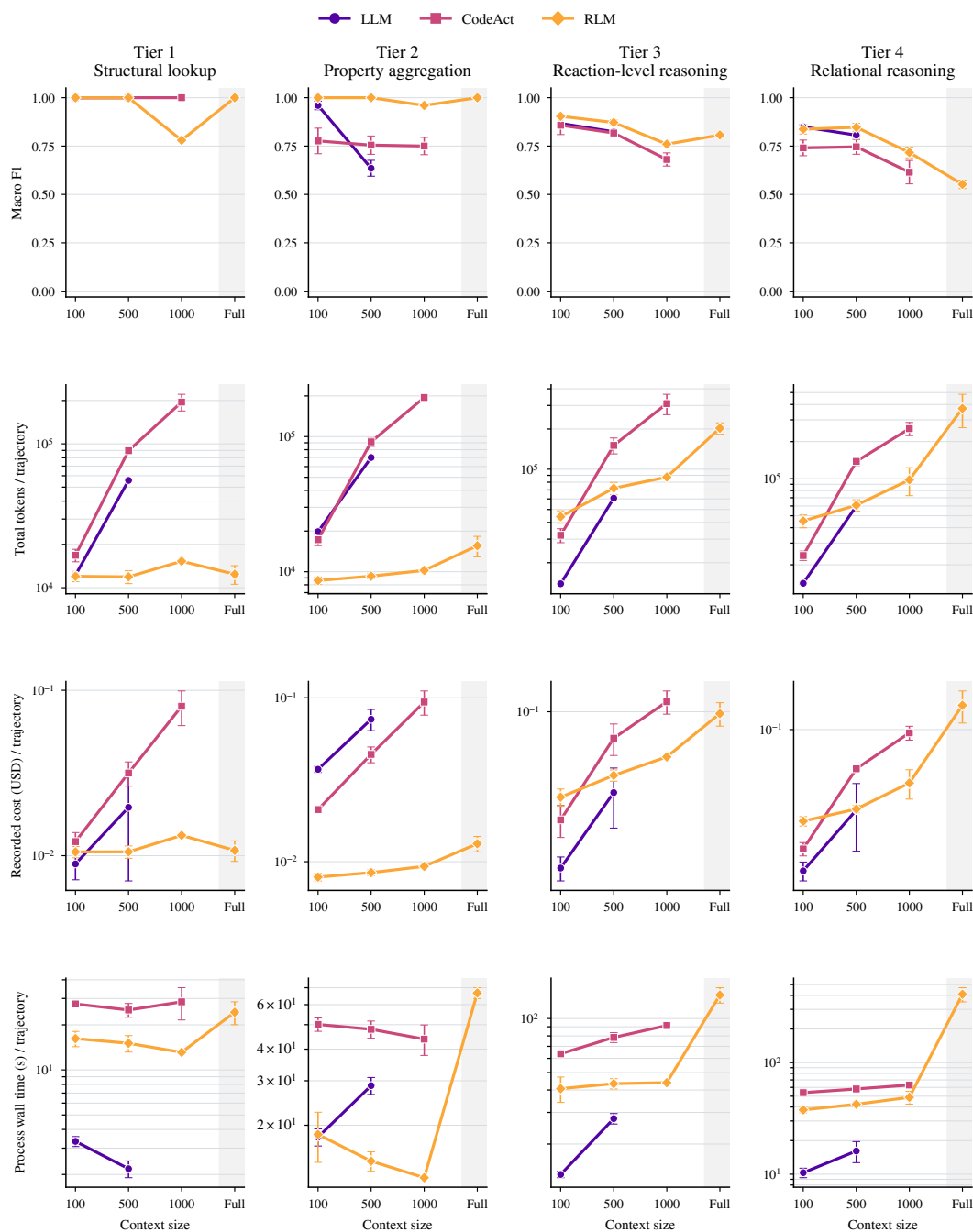


Figure 26: Gemini 3.7 Flash performance and efficiency by tier. Each row contains four tier panels. Bars show one population standard deviation across five repetitions. Cost is recorded billed cost in USD; token and process-wall-time means are normalized per successfully scored question trajectory. The $x = 1000$ extension covers CodeAct and RLM, while full-corpus access is RLM-only.

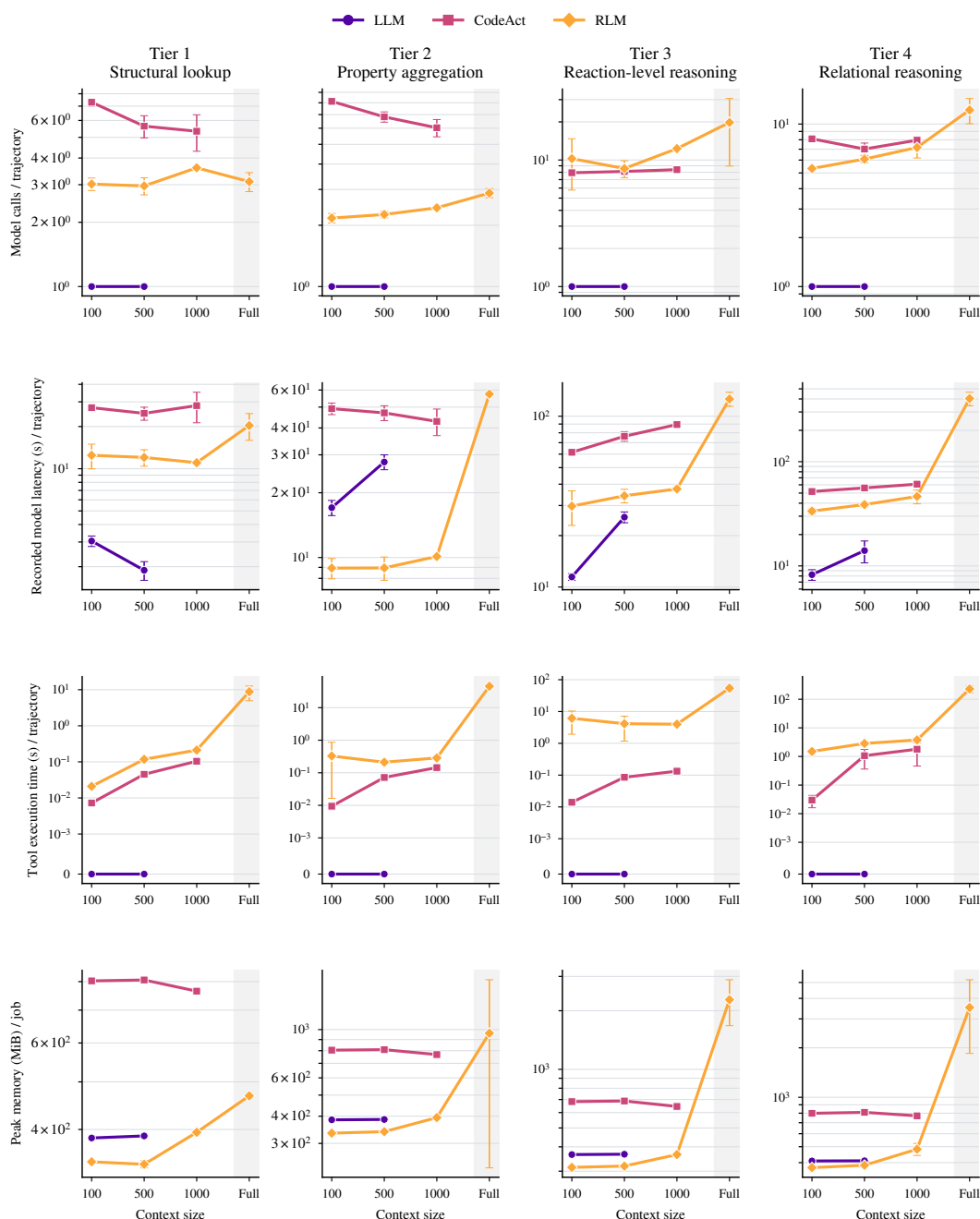


Figure 27: Gemini 3.7 Flash resource diagnostics by tier. Model calls count successful language-model API calls, recorded latency accumulates model-call latency, and tool time records local Python/RDKit execution. These additive measures are normalized per successfully scored question trajectory. Peak memory is the mean per-job maximum of the combined host process tree and Docker processes.

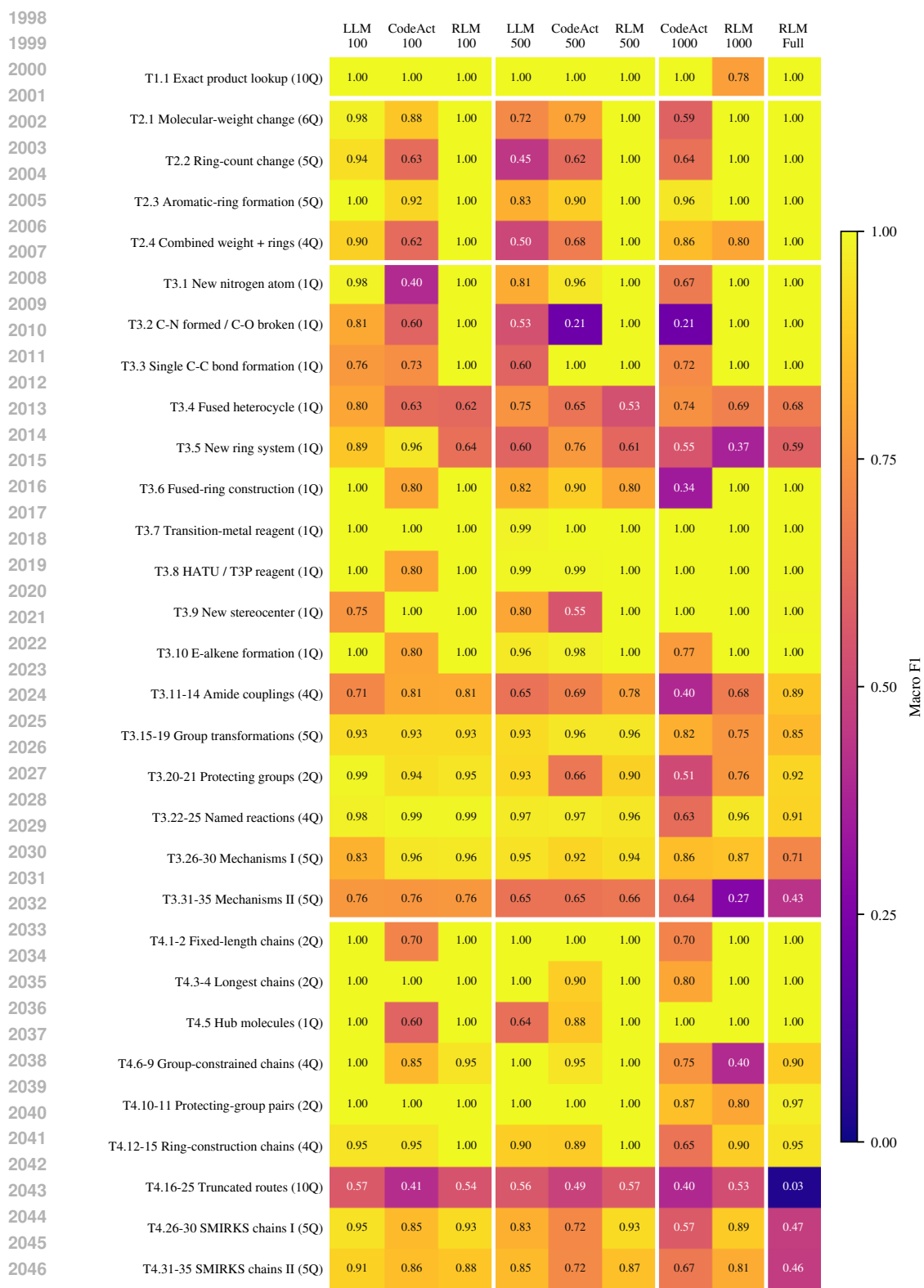


Figure 28: Gemini 3.7 Flash task-level macro-F1, averaged across five repetitions. Terminal failures contribute zero. The heatmap preserves all 30 task configurations and includes the $x = 1000$ CodeAct and RLM extensions; full-corpus access is RLM-only.

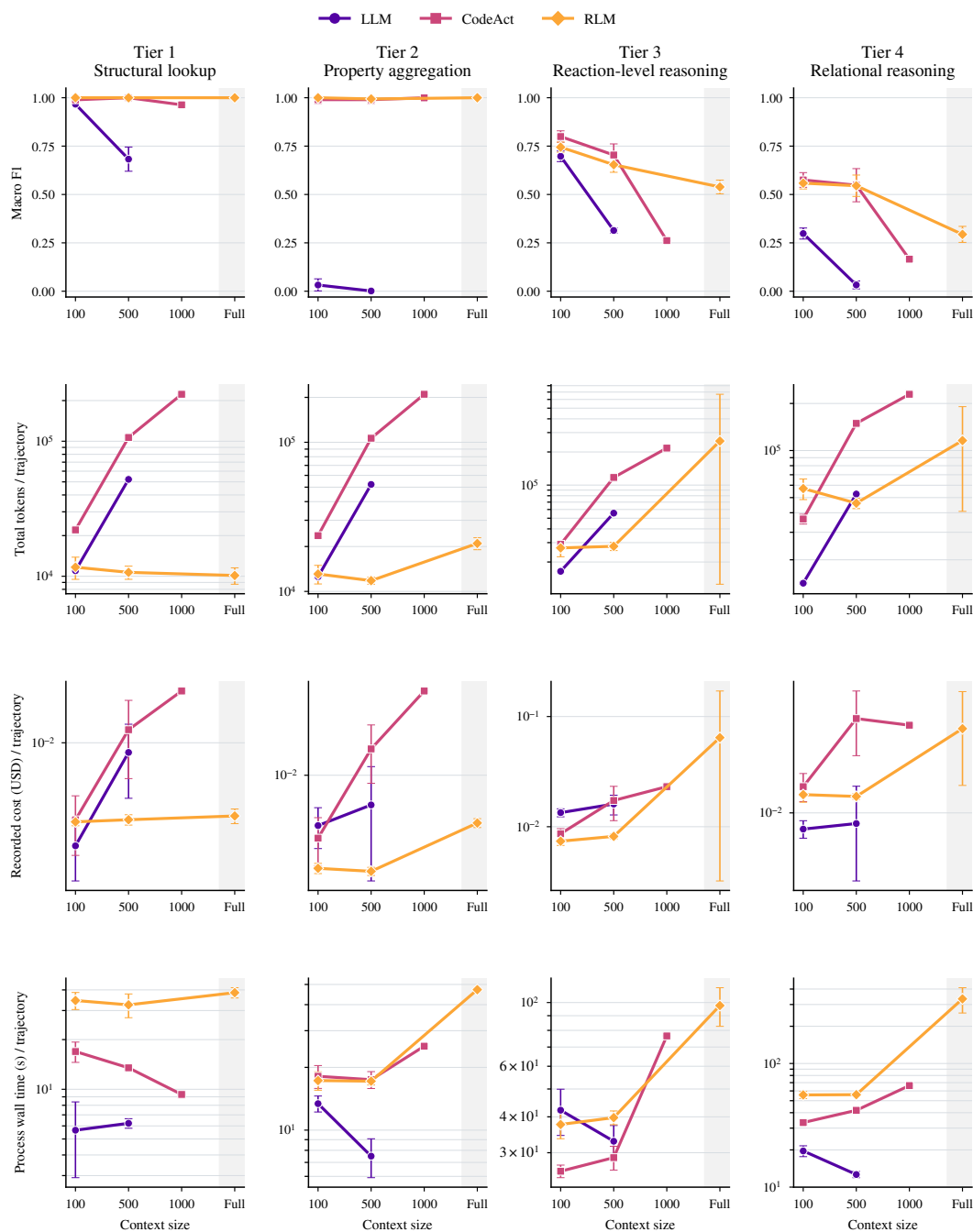


Figure 29: GPT-5 mini performance and efficiency by tier. Each row contains four tier panels. Bars show one population standard deviation across five repetitions. Cost is recorded billed cost in USD; token and process-wall-time means are normalized per successfully scored question trajectory. The $x = 1000$ extension covers CodeAct and RLM, while full-corpus access is RLM-only.

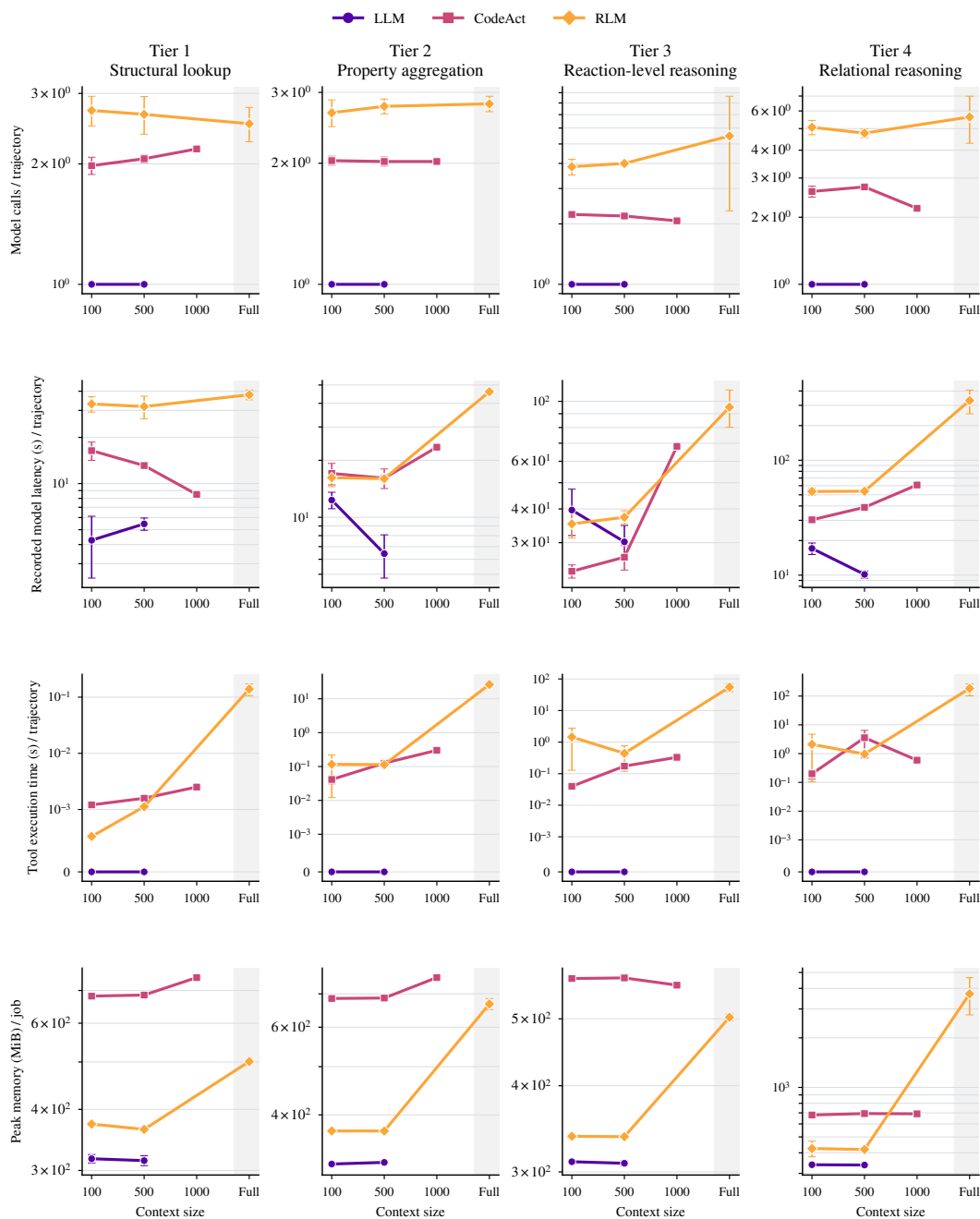


Figure 30: GPT-5 mini resource diagnostics by tier. Model calls count successful language-model API calls, recorded latency accumulates model-call latency, and tool time records local Python/RDKit execution. These additive measures are normalized per successfully scored question trajectory. Peak memory is the mean per-job maximum of the combined host process tree and Docker processes.

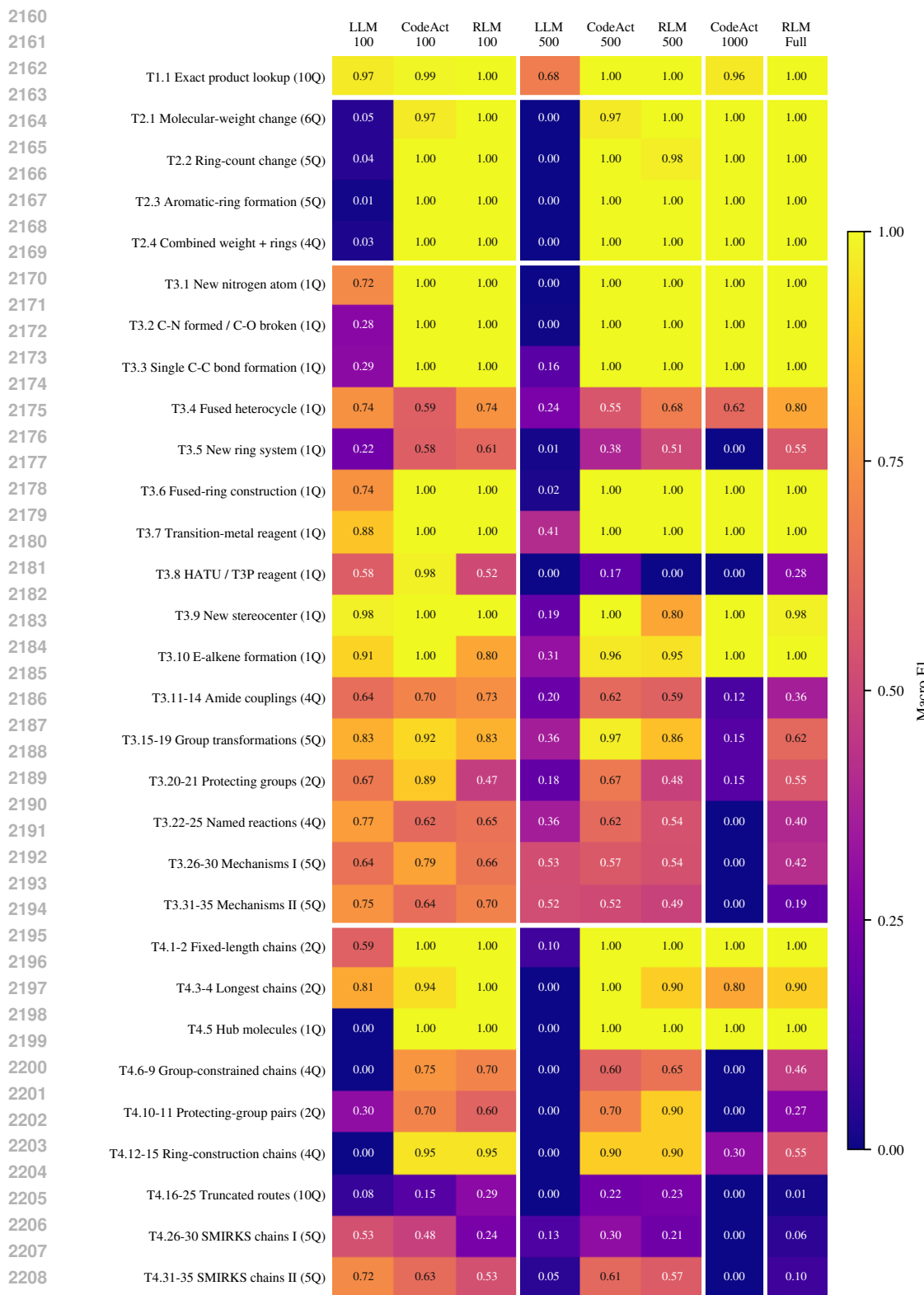


Figure 31: GPT-5 mini task-level macro-F1, averaged across five repetitions. Terminal failures contribute zero. The heatmap preserves all 30 task configurations and includes the $x = 1000$ CodeAct and RLM extensions; full-corpus access is RLM-only.

C FAILURE MODE ANALYSIS

This appendix documents chemistry-specific failure modes drawn from actual run logs, organized by tier and paradigm. Each entry names the run, the chemical error, and what the correct approach would have been. The ground truth for all tier-3 functional group and mechanism tasks is generated by applying the Rxn-INSIGHT SMIRKS templates and the multi-stage mechanism cascade from Joung (2025) via `AllChem.ReactionFromSmarts`; the ground truth for tier-4 chain tasks uses exact canonical-SMILES product-to-reactant linking.

C.1 TIER 3: FUNCTIONAL GROUP TRANSFORMATIONS (CODEACT)

C.1.1 RUNREACTANTS USED FOR TEMPLATE MATCHING (T3.11, RUN 20260420_215745)

Task. Retrieve all acyl chloride + primary amine amide coupling reactions. CodeAct generates a SMIRKS and calls `RunReactants` to test membership:

```
rxn = rdChemReactions.ReactionFromSmarts(smirks_to_use)
produced = rxn.RunReactants(reactant_mols)
# ...
if produced: # if any products produced
    matched = True
```

`RunReactants` asks whether the template can generate the observed product, a stricter and different question from “does this reaction belong to this class?” Any reaction whose leaving-group or atom-mapping representation deviates from the generated SMIRKS is rejected even when the chemistry is correct. Correct approach: `HasSubstructMatch` on separate reactant-side and product-side SMARTS. Result: acyl_chloride F1 = 0.0; macro F1 across 4 amide coupling variants = 0.23.

C.1.2 DECOUPLED AMINE-BOC TESTS (T3.16, RUN 20260618_181520, P = 0.113, R = 1.0, F1 = 0.204; 3,880 FALSE POSITIVES)

Task. Retrieve all reactions installing a Boc group on a primary amine. The model applies two independent tests, (i) any reactant nitrogen with `GetTotalNumHs() == 2` and (ii) any product fragment matching `N-C(=O)-O-C(C)(C)C`, with no atom-level linkage between them:

```
# Check for primary amine in reactants: N atom, neutral,
# with total H count == 2
has_primary_amine = False
for m in reactant_mols:
    for atom in m.GetAtoms():
        if atom.GetSymbol() == 'N' and atom.GetFormalCharge() == 0:
            if atom.GetTotalNumHs() == 2:
                has_primary_amine = True

# Check for Boc-on-N in products using substructure
has_boc_on_n = False
for m in product_mols:
    if m.HasSubstructMatch(pattern_boc_on_n):
        # pattern: "N-C(=O)-O-C(C)(C)C"
        has_boc_on_n = True
if has_boc_on_n and has_primary_amine:
    matches.append(idx)
```

Reactions where an unrelated NH₂ side chain coexists with a bystander Boc group satisfy both flags. Correct approach: identify the same nitrogen atom in both reactant and product by atom index, confirm its H-count drops from 2 to 1, and confirm the product nitrogen is bonded to a carbonyl carbon that is in turn bonded to a tert-butoxy oxygen, with all three checks tied to the same atom.

C.1.3 AMINE SCOPE RESTRICTION IN NAMED REACTION CLASSIFICATION (T3.22–T3.25, RUN 20260619_113011, MACRO F1 = 0.320)

Task. Classify Buchwald–Hartwig, Mitsunobu, Stille, and Suzuki reactions. The Buchwald–Hartwig classifier restricts to free primary aliphatic amines:

```
def count_primary_amines(mols):
    for a in m.GetAtoms():
        if a.GetSymbol()=='N' and a.GetFormalCharge()==0:
            # primary amine: two hydrogens and one heavy neighbor
            total_h = a.GetTotalNumHs()
            deg = a.GetDegree()
            if total_h==2 and deg==1:
                cnt += 1
# misses: piperazine (deg=2), Boc-NH (deg=2), ring N-H (deg=2)
```

Only 1 of 12 ground-truth reactions has a free benzylamine reactant; the remaining 11 involve secondary ring or protected amines (F1 = 0.154). Suzuki uses a product-minus-reactant halide count check that flags selective coupling on polyhalogenated arenes as false positives when one halogen survives (P = 0.333). Correct approach: use reaction-center SMARTS encoding the C(aryl)–N or C(aryl)–C(aryl) bond formation and match it against reactants and products together.

C.2 TIER 3: FUNCTIONAL GROUP TRANSFORMATIONS (RLM)

C.2.1 RECALL SATURATION FROM APPROXIMATE SMARTS (T3.15–T3.19, RUNS 20260618_172549 THROUGH 20260618_163048)

Task. Retrieve reactions matching each of five FG transformations (alcohol → azide, nitrile → amine, nitro → amine, alcohol → carboxylic acid, Grignard → tertiary alcohol). Across 5 runs at ctx = 500, the RLM achieves P = 1.0 but R = 0.2–0.4 per question, with 2 of 5 runs returning no matches at all (F1 = 0.0). The context forces 5 ground-truth reactions in; the model returns at most 1–2 correct ones. The RLM writes its own atom-level SMARTS to detect, for example, an alcohol oxygen connected to an sp³ carbon:

```
def find_azide_carbons(mol):
    # find azide triplet N1-N2-N3 (common SMILES N=[N+]=[N-])
    for a in mol.GetAtoms():
        if a.GetAtomicNum() == 7:
            for b in a.GetNeighbors():
                for c in b.GetNeighbors():
                    if c.GetIdx()==a.GetIdx():
                        continue
    # more systematic: iterate over all pairs of connected N atoms...
    # -- never returns results; loop body accumulates no results
```

The hand-written SMARTS omits part of the transformation’s atom environment, so it silently skips 3–4 of the 5 in-context positives. In 2 of 5 runs the code crashes or returns -1. Correct approach: derive reactant-side and product-side SMARTS from chemistry knowledge of each transformation (e.g., for alcohol-to-azide: sp³ carbon bearing an OH in the reactant; the same carbon bearing an organic azide N=[N+]=[N-] in the product), then apply HasSubstructMatch independently to the reactant and product molecules. This avoids fragile manual traversal of atom chains.

C.3 TIER 3: REACTION MECHANISM CLASSIFICATION (CODEACT)

C.3.1 SMARTS MISCALIBRATION (T3.26–T3.35, RUN 20260619_162404)

Task. Classify reactions by mechanism (ester hydrolysis, Wittig, Mitsunobu). CodeAct writes SMARTS at the wrong level of specificity in both directions:

```
# ester_hydrolysis: too broad -- matches any ester carbonyl
ester_smarts = Chem.MolFromSmarts("[CX3](=O)O[#6]")
# precision=0.3846, recall=1.00, F1=0.5556;
```

```

2322 # also catches transesterification
2323
2324 # wittig: requires formal charge [P+] adjacent to [C-]
2325 def has_phosphorus_ylide(mols, side_smiles_text):
2326     if atom.GetSymbol()=='P' and atom.GetFormalCharge() > 0:
2327         for nb in atom.GetNeighbors():
2328             if nb.GetSymbol()=='C' and nb.GetFormalCharge() < 0:
2329                 return True
2330 # precision=1.00, recall=0.20, F1=0.33; misses neutral phosphorane

```

For hydrolysis, the SMARTS must additionally require a hydroxide source ([OH⁻] or O) in the reactant field. For Wittig and Mitsunobu, the SMARTS should match the phosphine ylide C=P and azodicarboxylate N=N at the reaction center while allowing charge-state variants elsewhere in the molecule.

C.3.2 SUBSTRUCTURE-MAPPING FAILURE ON ALPHA-ALKYLATION (T3.31, RUN 20260620_021016, F1 = 0.0)

Task. Retrieve all keto alpha-alkylation reactions. The model requires an alkyl halide or sulfonate electrophile in the reagent pool, finds alpha-carbons (sp³ C adjacent to C=O with at least one H), then uses `GetSubstructMatches(reactant, product)` to establish an atom mapping and checks whether the mapped alpha-carbon gains a new carbon neighbor:

```

2343 if not has_alkyl_halide_or_sulfonate(reagent_pool):
2344     continue # rejects dimethyl sulfate, acrylates, etc.
2345 # ...
2346 matches_map = mol_p.GetSubstructMatches(query, useChirality=False)
2347 if not matches_map:
2348     continue # fails when product ring-closes or rearranges

```

The alkyl halide gate rejects non-halide electrophiles (dimethyl sulfate, Michael acceptors). The substructure-map gate fails when the product undergoes intramolecular cyclization (Dieckmann-type) because the product connectivity no longer contains the reactant as a subgraph. Both gates must pass simultaneously; their intersection is total failure. Correct approach: encode the two-step mechanistic logic directly. Check that the reactant has an enolizable carbonyl (sp³ alpha-C adjacent to C=O bearing at least one H), then verify by atom index that this alpha-carbon gains a new C-C or C-heteroatom bond in the product. This test accepts any electrophile class (halide, sulfonate, Michael acceptor) without relying on a fixed leaving-group list.

C.4 TIER 3: REACTION MECHANISM CLASSIFICATION (RLM)

C.4.1 MECHANISM MISIDENTIFICATION AT CONTEXT SIZE 500 (T3.26–T3.35, RUN 20260619_165857, MACRO F1 = 0.481)

Task. Identify reactions matching each of five mechanism templates (ester hydrolysis, Mitsunobu, Wittig, Knoevenagel, azide-alkyne cycloaddition). At `ctx = 500` the RLM processes all reactions in a single recursive call and stops after the most visually obvious matches. For ester hydrolysis (GT = 15) it predicts 1 reaction, incorrectly (F1 = 0.0): the base appears in the reagent field as [OH⁻] or [Na⁺] [OH⁻], so the RLM's reactant-only scan misses it. For Mitsunobu (recall = 0.2) and Knoevenagel (recall = 0.4) it identifies only the structurally most obvious examples and misses cases where the diagnostic groups (phosphine, activated methylene) appear in polyfunctional molecules. Correct approach: scan all three fields of each reaction SMILES; for cofactor-dependent mechanisms (ester hydrolysis, Wittig, Mitsunobu) check the reagent slot for the cofactor molecule.

C.4.2 MISSED MULTI-STAGE CASCADE ON BASE-CATALYZED TRANSESTERIFICATION (T3.32, RUN 20260620_021016, F1 = 0.0; 0 OF 5 IN-CONTEXT POSITIVES RECOVERED)

Task. Retrieve base-catalyzed transesterification reactions. The RLM extracts the acyl fragment from each reactant ester, compares it to the acyl fragment in each product ester, and checks for a negatively charged oxygen in the reagent pool:

```

def reagent_has_base(reagent_mols):
    # detect any negatively charged oxygen
    for atom in mol.GetAtoms():
        if atom.GetAtomicNum() == 8 and atom.GetFormalCharge() < 0:
            return True
    return False
# ...
if r_acyl == p_acyl: # PathToSubmol string compare, non-canonical
    if r_alk != p_alk:
        found_trans = True

```

The acyl fragment extraction via `FragmentOnBonds` produces non-canonical SMILES strings that differ between reactant and product even for identical acyl groups, so `r_acyl == p_acyl` returns False on all true positives. Independently, most transesterification reactions in the dataset use NaOMe, K₂CO₃, or Et₃N as base, none of which produce a bare [O-] in SMILES notation. Correct approach: canonicalize acyl fragments via `Chem.MolToSmiles` before comparing; detect base by a broader SMARTS covering alkoxide, carbonate, and tertiary-amine patterns; and verify the two structural hallmarks of transesterification (the acyl group R-C=O is conserved from reactant ester to product ester while the alkoxy group changes) without requiring a specific base representation.

C.5 TIER 4: 4-REACTION SYNTHESIS PREFIX PREDICTION (CODEACT AND RLM)

C.5.1 CODEACT: TARGET-AGNOSTIC N-H COMPLETABILITY TEST (T4.16–T4.25, RUN 20260701_133401, MACRO F1 = 0.311)

Task. Find all 4-reaction chain prefixes of withheld 5-step synthesis routes to 10 named target molecules. The prompt tells CodeAct the target name but withholds its SMILES; the final reaction step is absent from the context. CodeAct applies the same N-H test as a completability proxy for all 10 questions regardless of the target:

```

def has_acylatable_N(smiles):
    for a in m.GetAtoms():
        if a.GetSymbol() == 'N':
            # total number of hydrogens (explicit+implicit)
            if a.GetTotalNumHs() > 0:
                return True
    return False # true of ~80% of drug-like intermediates

```

The test is too broad for targets whose final step is N-acylation, and chemically irrelevant for targets where the final step is a C-C coupling, ring closure, or deprotection. Correct approach: use the target name to identify the final-step reaction class by SMIRKS, then match the terminal product of each candidate prefix against the reactant-side pattern of that SMIRKS via `HasSubstructMatch`.

C.5.2 RLM: LABEL-DERIVED STRING HEURISTICS (T4.16–T4.25, RUNS 20260701_020510 AND 20260701_034159, MACRO F1 = 0.24–0.36)

Task. Find all 4-reaction chain prefixes of withheld 5-step synthesis routes to 10 named targets. Without the withheld final-product SMILES, the RLM invents character-count heuristics from the question label. For “quinazoline halide”:

```

# >=2 halogen tokens (Cl/Br/F), >=2 'n' chars, contains 'c'
# -- matches pyrimidines, imidazoles, indoles equally well

```

For “phthalimide glutarimide” it looks for substring `OCc1cccc1` (Cbz) combined with `N1CCN` (piperazine). These SMILES string checks bypass substructure matching. Questions where the label maps to a structurally unique fingerprint (“uracil piperazine,” “benzamide pyrazole”) achieve F1 = 1.0; the remaining questions receive F1 = 0.0. Correct approach: enumerate candidate final reactions from the context by SMIRKS class and verify product-to-reactant connectivity directly.

C.6 TIER 4: 2-REACTION SMIRKS SEQUENTIAL CHAINS (CODEACT AND RLM)

C.6.1 CODEACT: REAGENT-HEURISTIC CHAIN CLASSIFICATION (T4.26–T4.35, RUN 20260630_130044)

Task. Find all 2-reaction chains $[r_0, r_1]$ where step 1 matches one fixed SMIRKS template, step 2 matches another, and a product of r_0 exactly equals a reactant of r_1 by canonical SMILES. For `boc_buchwald` (GT = 30 chains), CodeAct predicts 480 chains with F1 = 0.0; for `nitro_reduction_amide` (GT = 190), it predicts 0 chains. For Boc protection, the model builds candidate step-1 reactions by checking whether the reagent list contains the string Boc or OC(=O)OC(C)(C)C. This detects Boc-group transfers while omitting the mapped primary- or secondary-amine pattern on the reactant side of the SMIRKS. For step-2 Buchwald–Hartwig, Pd presence serves as the sole criterion, omitting the aryl-halide and amine reactants. The 480 false-positive chains arise because the product-to-reactant canonical SMILES linking step is absent: the model checks a set-intersection of product and reactant components using raw SMILES strings, so most pairs fail to link correctly and all 480 differ from the 30 ground-truth chains. For nitro reduction, the generated code detects “aryl bromination” (NBS, Br₂), although the task requires nitro reduction (Fe/AcOH, H₂/Pd); consequently, it produces no step-1 candidates.

```
# r0 (Boc): product contains substring "CC(C)(C)OC(=O)N"
if "CC(C)(C)OC(=O)N" in p or "OC(=O)N" in p and "C(C)(C)" in p:
    boc_rxs.add(idx)
# r1 (Buchwald): 'Pd' in reagent field
if 'Pd' in reag:
    buchwald_rxs.add(idx)
# linking: raw string set intersection (non-canonicalized SMILES)
prod_set = set(reactions[r0]['products'])
react_set = set(reactions[r1]['reactants'])
if prod_set & react_set:
    chains.append(f"{r0},{r1}")
# -- chemical-persistence condition: omitted entirely
```

Correct approach: implement each step classifier using structural hallmarks derivable from chemistry knowledge: Boc protection by confirming that a tert-butyl carbamate fragment (N-C(=O)-O-C(C)(C)C) appears in the product on a nitrogen that was unprotected in the reactant, and Buchwald–Hartwig by confirming a new C(aryl)–N bond forms between an aryl halide reactant and an amine. Then link consecutive reactions by comparing RDKit-canonicalized product SMILES of step 1 against reactant SMILES of step 2, and enforce persistence of the spine fragment across the chain.

C.6.2 RLM: CONTEXT-SCALE COLLAPSE (T4.26–T4.35, ACROSS 5 RUNS)

Task. Find all 2-reaction chains whose two steps match specified SMIRKS templates and satisfy linking and persistence constraints. At `ctx = 100` the RLM achieves macro F1 = 0.77 (5 runs); at `ctx = 500` F1 drops to 0.34; at full-dataset scale F1 = 0.03. At `ctx = 100` the RLM correctly writes SMIRKS-based step classifiers and applies canonical-SMILES linking. At `ctx = 500` the code becomes entangled: for `azide_staudinger` (GT = 2 chains in context), the model substitutes reagent-field checks for the required structural templates, using sodium azide for the first step and trivalent phosphine for the second. This reagent-name approach returns 116 false-positive chains vs. 2 GT. At full-dataset scale the recursive retrieval index returns only a handful of reactions per query, and the model abandons systematic enumeration in favor of opportunistic sampling.

```
# azide-forming step: reagents contain '[N-]=[N+]=[N-]'
# or 'Na' + azide substring
# Staudinger step: reagents contain 'P(' or 'P' followed
# by aromatic groups 'clcccccl'
# linking: azide product SMILES string exactly equals
# reactant component string
# -- SMIRKS templates: omitted
# -- RDKit canonicalization: omitted before comparison
```

Correct approach: implement step classifiers as structural chemistry rules (functional group appearance or disappearance detectable by `HasSubstructMatch`) that are independent of context size;

use canonical-SMILES equality as the sole product-to-reactant linking criterion, which is $O(n^2)$ but scale-invariant; and avoid reagent-name string matching entirely, as reagent naming is inconsistent across the dataset.

D HUMAN-BASELINE STUDY

Design. Three anonymous chemistry-aware volunteers completed stratified question subsets spanning all tiers and task families, with selected overlap for agreement measurement. Completing the full, broad suite is time-consuming, so the study used balanced chunks rather than asking each contributor to answer all 100 questions. Tools and AI assistance were allowed, while the annotator remained responsible for checking the method and final answer. The application hid the reference answer until the first submission, recorded active browser and self-reported offline time separately, autosaved work, and preserved subsequent revisions. The baseline analysis uses only first submitted, non-abstained answers; missing assignments are not counted as errors.

Rescoring and missing data. We preserved the checksum-verified raw exports and rescored them against canonical bundle `rxnhaystack-human-1.6.0` after the benchmark audit corrected affected reference sets. Set comparison is order invariant. Exact set match requires equality, while precision, recall, and F1 retain partial credit for large set-valued answers. One submitted abstention is excluded from accuracy but retained in completion and timing summaries. Post-reveal revisions cannot replace the independent baseline submission, and active and offline times are not summed because they may overlap.

The volunteers submitted 52 responses, of which 51 were non-abstained. They recorded 8.46 active browser hours, 10.71 browser wall hours, and 5.76 self-reported offline/tool hours in total; the latter can overlap with an open browser session. Median active time per submitted response increased from 3.32 minutes in Tier 1 to 7.44 minutes in Tier 4, while median offline time increased from 2 to 10 minutes (Figure 32). Mean confidence was 4.07 on a five-point scale. Tool-use records show RDKit on 51 responses, local scripts on 48, an LLM or coding agent on 45, offline processing on 13, literature or web search on 2, and consultation on 2; categories are non-exclusive.

The overlap yielded 11 pairwise reviewer–item comparisons across the three reviewer pairs. Raw correctness agreement ranged from 0.80 to 1.00 and pairwise Cohen’s κ from 0.545 to 1.000. Across all answered overlap, nominal Krippendorff’s α was 0.777 for submitted-answer identity and 0.744 for correctness. These agreement figures summarize the deliberately limited overlap and are interpreted alongside the item-level disagreement list rather than as precise population estimates. Machine-readable item metrics, timing records, tool-use counts, pairwise disagreements, and the immutable input checksums accompany the artifact.

Table 6: Human-baseline performance after corrected-ground-truth rescoring. Overall values macro-average the first submitted, non-abstained question answers.

Scope	Scored responses	Exact match	Precision	Recall	F1
Tier 1	7	1.000	1.000	1.000	1.000
Tier 2	12	0.833	0.854	0.854	0.854
Tier 3	16	0.500	0.929	0.951	0.925
Tier 4	16	0.625	0.695	0.741	0.680
Overall	51	0.686	0.848	0.869	0.842

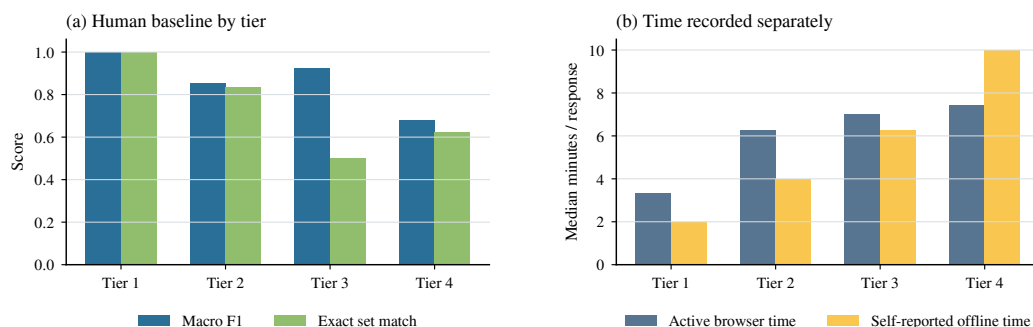


Figure 32: Human validation by tier. (a) Macro F1 and exact-set-match accuracy. (b) Median active browser time and self-reported offline/tool time per submitted response; the two timing measures are recorded separately and are not additive.

E BENCHMARK RELEASE AND CONTAMINATION POLICY

The anonymous supplement includes all 100 questions and reference answers, stable task metadata and prompts, answer-generation code, verification tests that regenerate the answer sets and check exact agreement with the released references, context builders and fixed sampling seeds, parsers and scorers, experiment definitions, and a version-pinned software environment. It also includes the chemistry rules and deterministic executor used in the controlled study. This documentation records provenance, composition, construction, and intended use following established dataset-documentation practice (Geburu et al., 2021). The code and benchmark will be released publicly as *RxnHaystack-Public-v1* upon acceptance.

The reaction corpus is reconstructed from the CC BY 4.0 *Reaction SMILES USPTO year 2023* dataset (van der Ling, 2023). The artifact records the upstream DOI and attribution, cleaning procedure, pinned RDKit version, row counts, and SHA-256 file fingerprints for the raw and cleaned corpora. Matching fingerprints verify byte-identical reconstruction. Dataset files remain outside the software repository and can be reconstructed with the supplied script.

The human-evaluation component comprises the annotation application, study definitions, instructions, assignment logic, schemas, and analysis code.

Public benchmark exposure can complicate later model comparisons, although the effect depends on the training regime and degree of exposure (Sainz et al., 2023; Bordt et al., 2025). For contamination-sensitive evaluation of future models, we will create a distinct *RxnHaystack-Heldout* stream from new targets, thresholds, reaction classes, task parameters, and sampling seeds selected after the public release. Its versioned instances and answers will remain private and be evaluated through a hosted service. Papers should identify the exact public version or held-out stream used, and the two scores will be reported separately.